

CHAPTER

3

PRINCIPLES OF DIGITAL DATA ACQUISITION AND FFT PROCESSING FOR SPECTRAL ANALYSIS

3.0 INTRODUCTION

In the previous chapter, one of the highlights was the FFT Programmable Data Collector. It is the current state-of-the-art choice for predictive maintenance programs to collect condition monitoring data. In addition, many of them have a built-in "Analysis Mode" which allows them to be used as an analyzer in the field. Since both data collectors and real time analyzers are also dynamic signal analyzers, it is important that the analyst fully comprehend the principles and techniques involved in digital data acquisition and FFT processing.

In this chapter, the basic properties of the Fast Fourier Transform will be presented. What happens when a finite, sampled record is transformed from the time domain will be considered to show the need to use filters. It will be shown that the length of the time record determines how closely the "filters" (or lines) will be spaced in the frequency domain, while the number of samples determines the number of "filters" in the frequency domain.

This chapter will highlight some undesirable characteristics, such as aliasing and leakage, and deal with solutions to these problems such as analog and digital filters and windowing. Some enhancement capabilities to make high resolution measurements, like "zoom" and averaging will likewise be featured.

Perhaps the highlight of this chapter will be the section which deals with the effects that the number of FFT lines and frequency span chosen have on the accuracy of the frequency. This is due to the fact that the more precise the displayed frequency, the more reliable will be the diagnosis of the source causing this frequency (a group of frequencies) to appear. Not only will this allow the analyst to more reliably diagnose the problem source, but also the problem severity (if a problem actually exists). Also, instruction will be given on how to improve frequency and amplitude accuracy by interpolation methods, and even choosing an instrument with sufficient dynamic range.

3.1 FFT PROPERTIES

The Fast Fourier Transform (FFT) is an algorithm for transforming the signal from the time domain to the frequency domain. However, the signal cannot be transformed to the frequency domain in a continuous manner. It must first be sampled and digitized. This means that samples from the time domain are digitized to samples in the frequency domain, as shown in Figure 1. Because of sampling, there is no longer an exact representation in either domain. However, a sampled representation can be closer to the ideal if the samples are placed closer together. Later in this section, the sampling spacing which is necessary to guarantee accurate results will be considered.

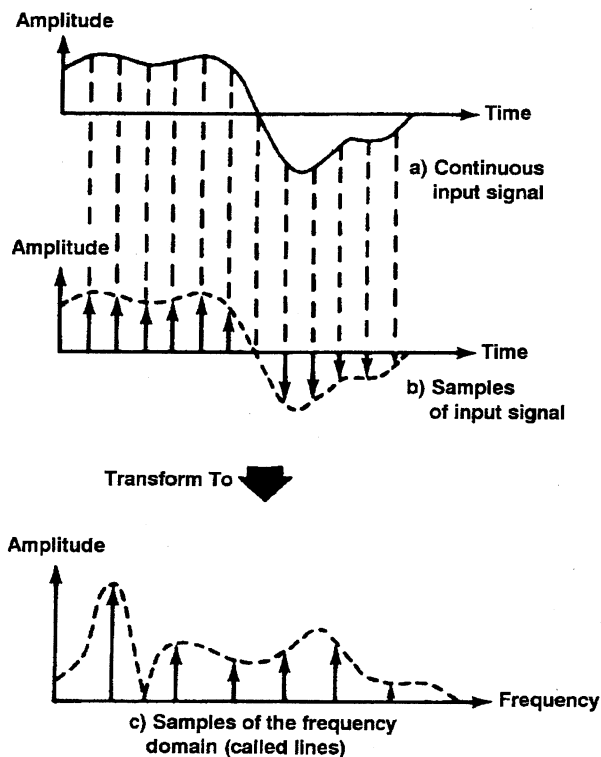


FIGURE 1
THE FFT SAMPLES IN BOTH THE TIME AND FREQUENCY DOMAINS (Ref. 1)

A time record is defined to be finite number of consecutive, equally-spaced samples of the input signal. Because it makes the transform algorithm simpler and much faster, the finite number of samples is restricted to a multiple of 2. Microprocessors work in powers of 2, called binary numbers. Thus, 1024 intervals or equally-spaced samples equal 2^{10} intervals (if an A/D instrument was specifically designed to do this, it would be referred to as a "10-BIT A/D INSTRUMENT").

This digitized time record is transformed as a complete block into a complete block of frequency lines. All the samples (of the input signal) of the time record are needed to compute each and every line in the frequency domain. This does not mean a single time domain sample transforms to exactly one frequency domain line. An explanation on this will be given in the next section.

Because the FFT transforms the entire time record as a total, there are no valid frequency domain results until the complete time record has been gathered. Once gathered, however, the oldest sample can be discarded, all the samples shifted in the time record, and a new sample added to the end of the time record as in Figure 2. Thus, once the time record is filled initially, there will be a new time record at every time domain sample. The point is, that after the initial time span, there will be rapid changes in the spectra. Although this is not practical, it could be utilized.

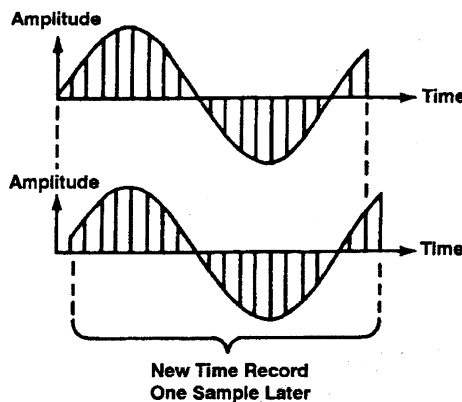


FIGURE 2
A NEW TIME RECORD ADDED EVERY SAMPLE AFTER THE TIME
RECORD IS FILLED (Ref. 1)

3.11 HOW MANY SPECTRAL LINES ARE THERE?

The FFT algorithm is a "complex-valued" operation; that is, it produces a "real" and an "imaginary" result, and the components of the frequency domain will appear at both positive and negative frequencies. The complex plane is 0 (inclusive) to 2π . The positive vector lies in the upper half of the complex plane. So a positive frequency is located between 0 and π . The negative vector lies in the lower half of the complex plane. Thus, a negative frequency is located between π and 2π (remember there are 2π radians per revolution of a rotating shaft).

For real-valued signals, it can be shown that the components with positive frequencies are mirror images of their corresponding components with negative frequencies. Therefore, it is common practice to discard the redundant, negative frequency values when graphing the amplitude of the FFT of a real-valued signal. Since only half of the information returned from the FFT algorithm is displayed, component amplitudes are multiplied by two before being displayed (except for the DC component, which is at zero frequency on the real axis).

In summary, each of these components in the frequency domain is complex-valued; that is, they each have both amplitude and phase. Thus, the FFT transforms a finite number of equally-spaced samples from the time domain to only half as many lines in the frequency domain (see Equation 1). The reason for this is the same as stated above - each frequency line actually contains two pieces of information - amplitude and phase.

Thus, for a real-valued signal (the numbers are not complex), 800 data points would be required to compute a 400 line spectrum; 1600 data points for an 800 line spectrum, etc. However, some of the high frequency data is "discarded" to compensate for the roll-off of the anti-aliasing filters. In the "real world", 1024 data points are actually required for a 400-line transform; 2048 points for an 800-line transform, etc.

3.12 WHAT IS THE SPACING OF THE LINES?

The lowest frequency that can be resolved must be based on the length of the time record. If the period (T) of the input signal is longer than the time record, there is no way of determining the period. Therefore, the lowest frequency line of the FFT must occur at a frequency equal to the reciprocal of the time record length.

$$F = \frac{1}{T} \quad \text{where: } F = \text{Frequency (Hz or CPM)} \\ T = \text{Period (sec or min)}$$

In addition, there is a frequency line at zero Hz-DC. This is merely the average of the input over the time record. Although it has no practical value, it does help to establish that the spacing between these two lines (and hence every line) is the reciprocal of the time record. Thus, the entire spectral display is made up of many individual vertical lines (or "bins") located adjacent to one another along the frequency axis. Only bins with amplitude information in them will display an FFT peak.

3.13 WHAT IS THE FREQUENCY RANGE OF THE FFT?

The highest frequency which can be measured is:

$$F_{MAX} = \frac{\text{SAMPLE SIZE}}{2} \times \frac{1}{\text{PERIOD OF TIME RECORD}} \quad \text{Eqn. 1}$$

because there are only 1/2 as many lines in the frequency domain, spaced by the reciprocal of the time record starting at zero Hz. The usefulness of this frequency range can be limited by a problem called "aliasing" which will be discussed in a later section.

The number of time samples (sample size) is fixed by the implementation of the FFT algorithm. Therefore, the period of the time record (sec/cycle) must be varied to change F_{MAX} (cycles/sec). To do this, the sample rate must be varied so that there always is the chosen, fixed number of time samples in the variable time record period. To cover high frequencies, the time record period must be shorter so that sampling is very fast.

3.2 SAMPLING AND DIGITIZING

Recall that the input is a continuous analog voltage coming from the accelerometer and is proportional to the acceleration. Since the FFT requires digitized samples of the input for its digital calculations, a "sampler" and an "analog to digital converter" (A/D) needs to be added to the FFT processor to create a spectrum analyzer.

For the analyzer to have the high accuracy needed, the sampler and the A/D converter must be quite good. The A/D converter must have high resolution and linearity. For 70 dB of dynamic range, the A/D converter must have at least 12 bits of resolution (a "12-bit A/D instrument has 4096 equally-spaced samples or intervals in the display, equal to 2^{12} intervals). Typically, the A/D converter must be able to acquire at least a hundred thousand readings per second.

3.3 ALIASING

The reason an FFT spectrum analyzer needs so many samples per second is to avoid "aliasing". Aliasing is not always bad. It is called "mixing" or "heterodyning" in analog electronics, and is commonly used for tuning household radios and televisions.

3.31 ALIASING IN THE FREQUENCY DOMAIN

It is easy to see that a sampling frequency that is exactly twice the input frequency would not always be enough in the time domain. If the sampling rate is low (<2), the alias products will fall in the frequency range of the input ("creating" fictitious frequencies) and no amount of filtering will be able to remove them from the signal. However, as shown in Figure 3, if we sample at greater than twice the highest frequency of our input, the alias products will not fall within the frequency range of our interest. Therefore, a filter (or the FFT processor which acts like a filter) which is placed after the sampler will remove the alias products, but pass the desired input signals.

Two signals are said to alias if the difference of their frequencies falls within the frequency range of interest. This difference frequency is always generated in the process of sampling.

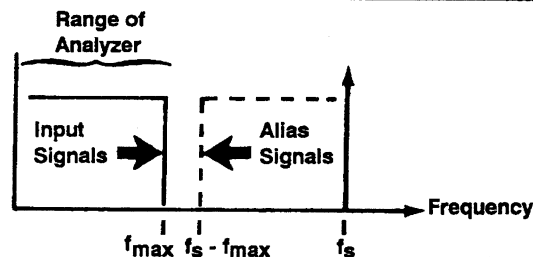


FIGURE 3

A FREQUENCY DOMAIN VIEW OF HOW TO AVOID ALIASING (Ref. 1)

3.32 THE NEED FOR AN ANTI-ALIAS FILTER

The only way to be certain that the input frequency range is limited is to add a low-pass filter before the sampler and the A/D converter. Such a filter is called an anti-alias filter. The gradual roll-off area of the filter is known as the transition band. Large input signals are not well attenuated in the transition band; they can still alias. To avoid this, the sampling "frequency" is raised to twice the highest frequency of the transition band. Typically, this means that to do this the sample rate actually is now 2.5 to 4 times the maximum desired input frequency.

3.33 THE NEED FOR MORE THAN ONE ANTI-ALIAS FILTER

To reduce the frequency span, the sample rate and the anti-alias filter frequency must be reduced by the same amount. To cover a wide range of frequencies, there needs to be many anti-alias filters (at least 12) for each channel with this setup. Unfortunately, they are very expensive, particularly the lowest frequency filters. Also, a different filter has to be utilized every time the sample rate of the A/D converter is changed. These are analog filters.

3.34 DIGITAL FILTERING

A digital filter filters the input signal after it has been sampled and digitized. The A/D sample rate is set to the rate needed for the highest frequency span. Rather than changing to a different anti-alias filter each time, a single analog anti-alias filter is used. Then, a digital filter is added after the A/D converter. It not only filters the digital representation of the signal to the desired frequency span, but it also reduces the sample rate to the needed rate of the frequency span.

3.35 FORMULAS USED TO CALCULATE t_{MAX} AND F_{MAX}

Nyquist's Sampling Theorem, briefly stated, is sampling must be done at a frequency rate of at least twice the highest frequency component of interest in order to not lose any information contained in the sampled signal. Thus, to satisfy Nyquist's Theorem, select a sampling rate that is slightly higher than twice F_{MAX} ($2.56 \times F_{MAX}$ is typical). The following are conversion formulas to calculate the time required to gather data if the F_{MAX} is known and vice versa.

where:

$$t_{MAX} = \frac{(60)(\#FFT\ Lines)}{FREQ.SPAN} = \frac{(60)(SAMPLE\ SIZE)}{(2.56)(FREQ.SPAN)} \quad \text{Eqn. 2}$$

- t_{MAX} = Total Sampling Period setting how often amplitude is measured and stored (sec).
SAMPLE SIZE = Number of Analog to Digital Conversions to be used to Construct the Time Waveform (Samples - most often 1024 Samples to provide a 400 line FFT).
FREQ. SPAN = Frequency Span from F_{MIN} to F_{MAX} (CPM); which often is from 0 - F_{MAX} .

where:

$$F_{MAX} = \frac{(60)(\#FFT\ Lines)}{t_{MAX}} = \frac{(60)(SAMPLE\ SIZE)}{(2.56)(t_{MAX})} \quad \text{Eqn. 3}$$

- t_{MAX} = Total Sampling Period setting how often amplitude is measured and stored (sec).
SAMPLE SIZE = Number of Analog to Digital Conversions to be used to Construct the Time Waveform (Samples - most often 1024 Samples to provide a 400 line FFT).
 F_{MAX} = Maximum Spectral Frequency or Frequency Span (CPM)

If this condition is violated (by not using an anti-aliasing filter), higher frequencies will be represented as lower frequencies in the sampled waveform. An example of this phenomenon may be seen in old western movies in which the wagon wheels appear to rotate backwards; this is because the film speed (sampling rate) was too slow compared to the speed of the wagon wheels. Vibration analysts can observe this same effect if they vary the frequency of a strobe light that is flashing on a part that is rotating at constant speed.

3.4 WINDOW SELECTION

3.41 THE NEED FOR WINDOWING

If a sine wave is periodic in the time record, its frequency spectrum would be a single FFT line if no windowing was used. However, if a sine wave is not periodic in the time record, its power will be spread throughout the spectrum. This smearing of energy throughout the frequency spectrum domain is known as "leakage". Energy leaks out of one resolution line of the FFT into all other lines as shown in Figure 4.

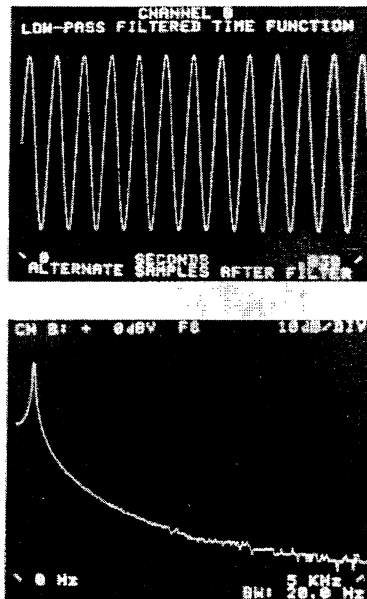


FIGURE 4

LEAKAGE OCCURS WHEN SINE WAVES ARE NOT PERIODIC IN THE TIME RECORD, AND AFFECTS THE REMAINDER OF THE SPECTRUM (Ref. 1)

It is obvious here that the leakage problem can entirely mask small signals located close to the sine wave(s). The solution to this problem is known as "windowing".

Notice that most of the problem seems to be at the edges of the time records; the center portion is a good sine wave. If the FFT could be made to ignore the ends and concentrate on the middle of the time record, the spectrum in the frequency domain should more closely produce a single peak. Windowing accomplishes this task - it forces the sampled data at the beginning and at the end of the sampling period to be equal to zero, thereby minimizing leakage.

3.42 WHAT IS WINDOWING?

If the time record was multiplied by a function that is zero at the ends of the time record and large in the middle, the FFT could be concentrated on the center portion of time record. One such function is shown in Figure 5A(c). Such functions are called "window functions" because they force us to view data through a narrow window. Figure 5A(b) shows the discontinuity in the time record. By applying a (HANNING) window, as in Figure 5A(c), the discontinuity is "filled in" so that the sample signal appears to be continuous, as shown in Figure 5A(d). Thus, the sampled data is forced to start and end at zero. Figure 5B is a simplified illustration of a one-cycle waveform in which the time record is not periodic. With Figure 5B(b) a (HANNING) window applied, the non-periodic waveform is forced to "appear" to be "periodic".

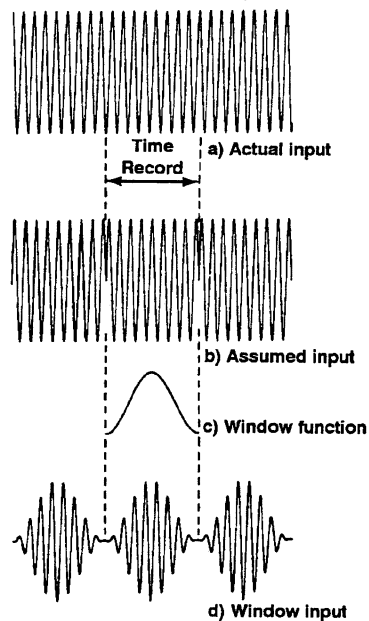


FIGURE 5A
THE EFFECT OF WINDOWING IN THE TIME DOMAIN (Ref. 1)

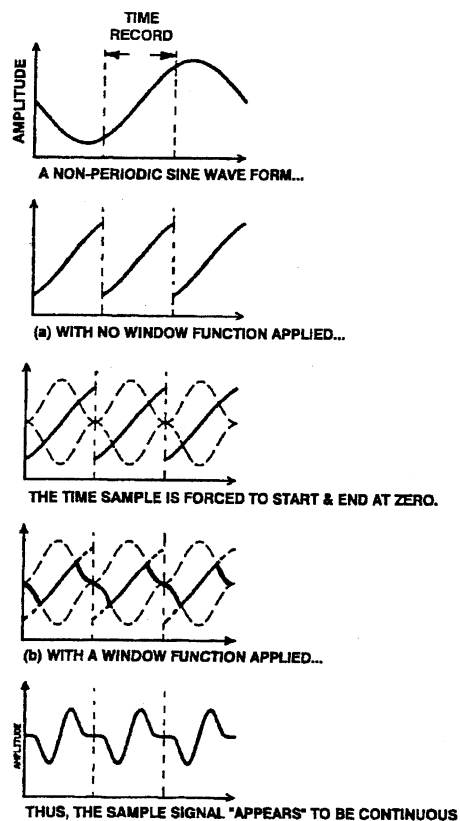


FIGURE 5B
A SIMPLIFIED ILLUSTRATION TO SHOW HOW WINDOWING FORCES A NON-PERIODIC WAVEFORM TO "APPEAR" TO BE PERIODIC (Ref. 3)

However, it is important to realize the input data has been tampered with by windowing and perfect results cannot be expected. The FFT assumes the input looks like figure 5A(d), something like an amplitude-modulated sine wave. This will produce a frequency spectrum that will be closer to a single line peak than one would be from Figure 5A(b), but it is still not 100% correct. Although Figure 6 shows a vast improvement from windowing data which is not periodic in the time record, it may still not be correct in that spectral components (or sidebands) might be "produced", but none truly exist in reality. Figure 7 demonstrates that the windowed data does not have as narrow a spectrum as an unwindowed function which is periodic in the time period.

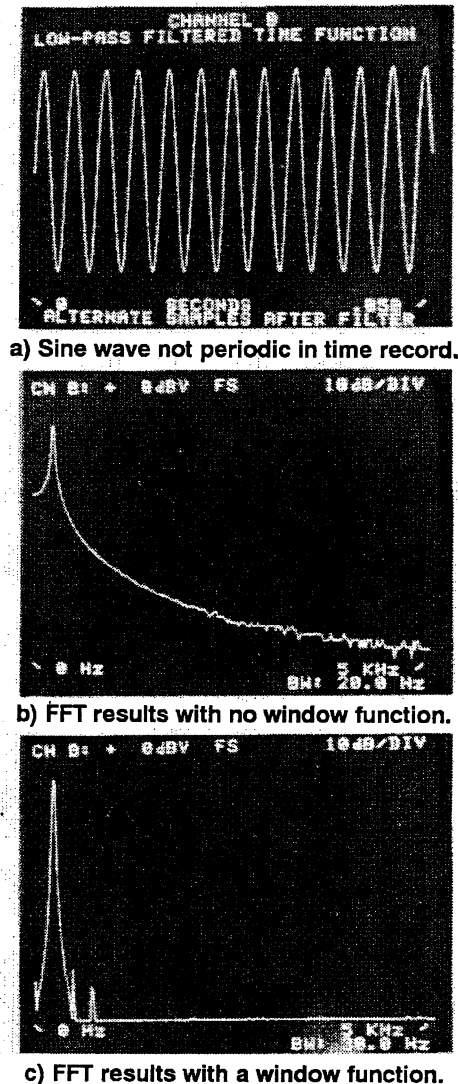
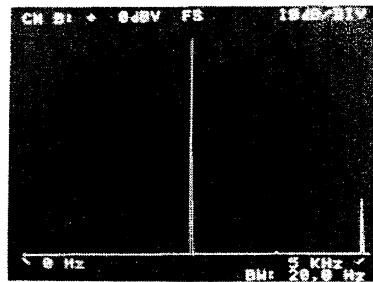
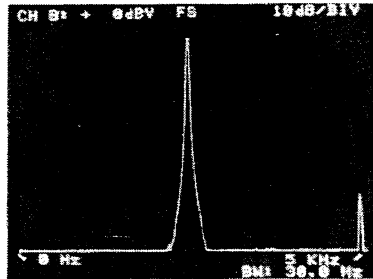


FIGURE 6
LEAKAGE REDUCTION WITH WINDOWING (Ref. 1)



a) Leakage free measurement - input periodic in time record



b) Windowed measurement - input not periodic in time record

FIGURE 7

WINDOWING REDUCES LEAKAGE, BUT DOES NOT ELIMINATE IT (Ref. 1)

3.43 THE HANNING WINDOW

The most commonly used window in vibration analysis is the Hanning window. In fact, the window used in Figure 5 was a Hanning window. The Hanning window does a good job with sine waves, both periodic and non-periodic. However, even with the Hanning window, some leakage will be present when the signal in the time record is not periodic.

Recall that the FFT acts like a set of parallel filters. Figure 8 shows the shape of those filters when Hanning windows are used. Notice that the Hanning function gives the filter a very rounded top. If a component of the input signal is centered in the filter, its amplitude will be measured accurately. In fact, it will be periodic in the time record. On the other hand, the filter shape will attenuate the component by up to 1.5 dB (-16%) when it falls midway between the filters.

To illustrate, using Figure 8 consider the following example:

Induction Motor Speed = 1785 RPM
 F_{MAX} = 12,000 CPM
 #FFT Lines = 400 lines (Hanning Window)
 Freq. Resolution = 30 CPM/line (1785 RPM will fall midway between 1770 and 1800 CPM bins meaning its amplitude reading is 16% lower than actual amplitude).
 If F_{MAX} = 6,000 CPM, 1785 RPM would fall in the middle of a bin, meaning there would be no reduction in the amplitude reading.

Thus, with the Hanning Window, amplitude measurements are always +0% to -16% of the actual.

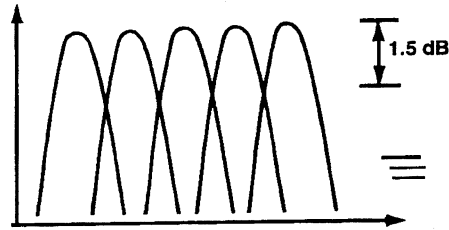


FIGURE 8
HANNING WINDOW PASSBAND SHAPES (Ref. 1)

That is, while the Hanning window provides improved frequency resolution, it sacrifices amplitude accuracy. Therefore, Hanning windows will lose information from transient events such as the impact test shown in Figure 9a. If the transient input signal is multiplied by the Hanning window function in Figure 9b, it would produce the highly distorted signal shown in Figure 9c. The frequency spectrum of an actual transient with and without the Hanning window is shown in Figure 10. The Hanning window has taken the transient, which naturally has energy spread widely through the frequency domain and made it look more like a sine wave.

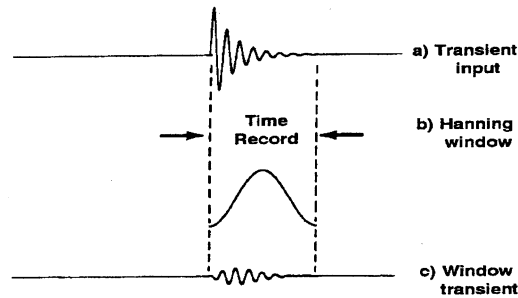


FIGURE 9
A HANNING WINDOW LOSES INFORMATION FROM TRANSIENT EVENTS (Ref. 1)

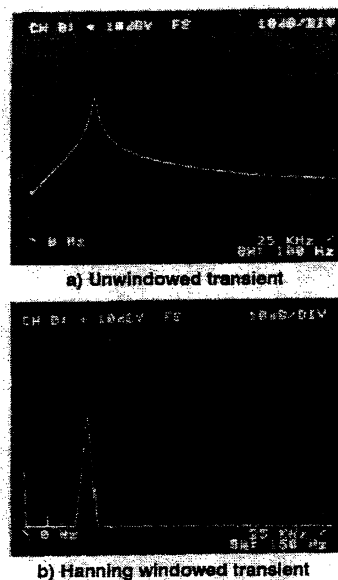


FIGURE 10
SPECTRA OF TRANSIENTS WITH AND WITHOUT A HANNING WINDOW (Ref. 1)

3.44 THE UNIFORM (RECTANGULAR) WINDOW

Since Hanning windows do not work well with transient vibrations, a uniform window (no window) could be used because it weights all of the time record uniformly. In the case of a transient, the analyzer can be set up so that there is no need for windowing the input because the transient occurs completely within the time record. This is called a "self-windowing function". It generates no leakage in FFT and so needs no window. Uniform windows are very useful for impact natural frequency tests. On the other hand, if a uniform window is used on a non-periodic sine wave, it will not eliminate leakage or wide skirts in the spectrum. In addition, amplitude variation for a uniform window can be up to 36%.

3.45 THE FLAT TOP WINDOW

Recall that the Hanning function gives the filter a very rounded top. Although this characteristic is desirable to clearly identify frequency peaks, it is unacceptable if the signal's amplitude needs to be accurately measured. The solution is to provide a window function that gives the filter a flatter passband, as shown in Figure 11.

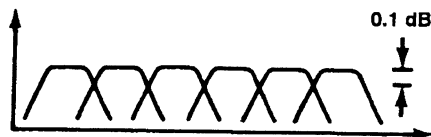


FIGURE 11
FLAT TOP PASSBAND SHAPES (Ref. 1)

The amplitude error from this window function does not exceed .1 dB (1%). The accuracy improvement does not come without its price, however. Figure 12 shows that the top of the passband has been flattened at the expense of widening the skirts of the filter. Therefore, some of the ability to resolve a small component, closely spaced to a large one, is lost.

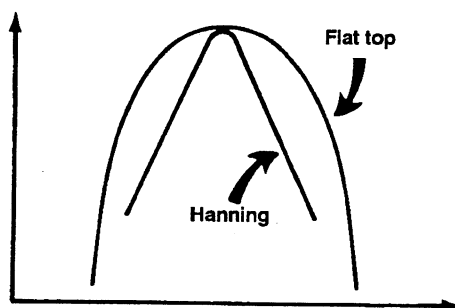


FIGURE 12
COMPARISON OF HANNING TO FLAT TOP WINDOW SHOWS REDUCED FREQUENCY RESOLUTION CAPABILITY OF THE FLAT TOP WINDOW (Ref. 1)

Many other window functions are possible but the three listed above are by far the most common for general measurements. In summary, the Flat Top window provides the best amplitude measurement, the Kaiser-Bessel window provides the best frequency separation, whereas the Hanning window provides the best compromise.

3.5 AVERAGING

The standard technique in statistics to improve the estimates of a value is to average. Two kinds of averaging are usually available, RMS (or "power" averaging) and synchronous time averaging.

3.51 RMS (POWER) AVERAGING

RMS averaging can be thought of as amplitude averaging. Each frequency's component in a power spectrum is proportional to the square of the corresponding component in the Fourier Transform. For the purpose of emphasizing the amplitude of the FFT, the analyzer will compute the power spectrum first; then take the square root of each component.

Note two practical consequences: the power spectrum requires less calculation time than the Fourier Transform because the square root calculations are not performed. Thus, any contrast between the high and low amplitudes is accentuated because the quantities are squared. Naturally, the Power Spectrum does not provide any phase information at all; only the differences in amplitude are emphasized. What we are trying to determine is, the average (or RMS) amplitude of the signal.

This averaging technique is very valuable for determining the average (RMS) power in any of the filters of an analyzer. The more averages that are taken, the better the estimate of the power level, making it more and more accurate.

If a small signal is measured in the presence of noise, the signal-to-noise ratio will not be improved with RMS averaging. RMS averaging does not "filter out" the noise; it includes the noise. Thus, the accuracy of the total power (signal-plus-noise) will be improved with RMS averaging, but not the signal-to-noise ratio. It does nothing to reduce unwanted background noise.

RMS averaging should be used when spectral-component-amplitude levels vary significantly. The result of an RMS average of successive FFT's is an improved estimate of the mean level of the amplitude. It causes varying amplitudes to converge to their mean value.

3.52 LINEAR AVERAGING AND SYNCHRONOUS TIME AVERAGING

The major difference between linear and synchronous time averaging is that a "triggering device" is needed for a synchronous time average, but is not necessary for a linear average. The triggering device is used to provide a synchronizing signal; the most commonly used devices are a photocell or an electromagnetic pickup.

Unlike power averaging, linear averaging does improve the signal-to-noise ratio. In many cases, the amplitudes of fundamental and passing frequencies are fairly constant, but is accompanied by some random amplitudes and frequencies. Even in cases where the amplitudes are low, the discrete frequency amplitudes remain fairly constant, while the random signal amplitudes will fluctuate between zero and some peak value. As a result, the average amplitudes of the random signals will be significantly less than their peak values. In both cases, the amplitudes that are stable remain close to the same value, regardless of the number of averages taken. On the other hand, random or transient amplitudes tend to approach zero as more and more averages are taken. Figure 13 shows a time record buried in noise. After taking 128 averages in this case, the resulting time record and frequency spectrum shows a marked improvement in the signal-to-noise ratio.

Linear averaging is a useful method to obtain repeatable data for fault-trending in PMP's. Usually 4, 8, or 16 averages are specified in the program software. That is, if it desired to average 16 samples, the "averager" will automatically proceed to sample the data 16 times, each time representing 1/16 of the data present. After the 16th sample has been completed, the averaged

spectrum, retained in memory, is displayed. Since the vibration sensed at each measurement point is the total vibration from the whole environment, the disadvantage of linear averaging is that it does not differentiate between the vibrations that only originate from the point and those that are only "sensed" at that point.

On the other hand, synchronous time averaging can be used to differentiate between vibration signals originating directly from the machine component of interest and eliminating those from nearby machines or any nonsynchronous components. In this case, a synchronizing signal is used to "trigger" the start of the time record. In this manner, a sample of the vibration is taken at the same moment with respect to shaft rotation each time. Therefore, the periodic part of the input where the trigger is targeted will always remain the same in each time record taken. If a series of these triggered time records are added together and divided by the number of records, N, taken, the synchronous time average can be computed.

Only harmonic vibrations from the machine (or shaft) being targeted by the trigger (photocell, strobe, laser tach, etc.) are emphasized by synchronous time averaging. The nonsynchronous frequencies will not be displayed because they are not integer multiples of the 1X RPM synchronizing signal. Thus, bearing defect frequencies, electrical line frequencies and multiples, as well as other harmonically distorted signals (even 1X RPM multiples from a nearby shaft), will be forced to zero (even if the speed difference is small like 1 or 2 RPM) if many averages are taken. This method is very useful for analyzing equipment that have many components rotating at nearly the same speed, such as a bank of fans, rolls on a paper machine, or on gears and shafts in a gearbox. After both results are transformed to the frequency domain, the running speed and harmonics are clearly seen and can be accurately measured because of the reduced noise floor (signal-to-noise ratio is enhanced).

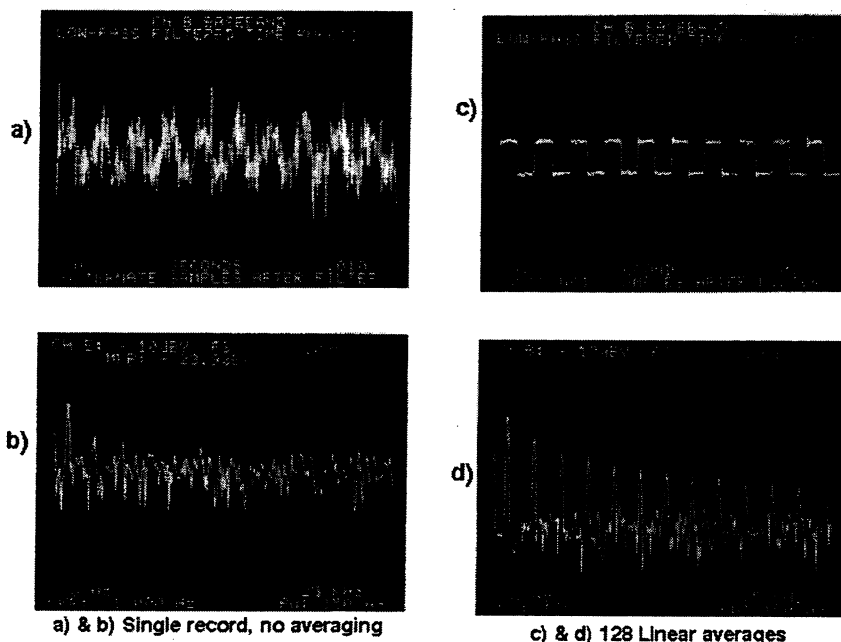


FIGURE 13
SYNCHRONOUS AVERAGING REDUCES THE NOISE FLOOR AND
ALLOWS THE HARMONICS TO BE EMPHASIZED (Ref. 1)

3.6 OVERLAP PROCESSING

Recall that the time record is not constant but is deliberately varied to change the frequency span of the analyzer. For wide frequency spans the time record is shorter. That is, first it will take a certain amount of time to capture the first time record according to the laws of physics, no matter whose analyzer one might have and whether or not they have a 32-bit DSP chip installed or not. The required time for this first block is given by Equation (2) which is repeated here for the reader:

$$t_{MAX} = \frac{(60)(\# \text{ FFT Lines})}{\text{FREQ.SPAN}} = \frac{(60)(\text{SAMPLE SIZE})}{(2.56)(\text{FREQ.SPAN})} \quad \text{Eqn. 2 (Repeated)}$$

After capturing the time record, it will then require a finite time to compute the FFT (this is where the high speed processors have an effect).

This fact cannot be overlooked. It will be especially important when a low frequency analysis needs to be made. In this case, the gathering of a time record requires much longer than the FFT computation time. Thus, the FFT processor sits idle much of the time. This can be seen in Figure 14 where the time record and the FFT are represented as "blocks" of time. If, instead of waiting on an entirely new time record, the new record was "overlapped" with some of the old data, a new spectrum could be obtained as often as the FFT was computed (after capturing the first block as per Equation 2). This is "Overlap Processing" and is illustrated in Figure 15. Notice that an FFT is obtained at the end of each time record. The time (space) between each new FFT represents the time it requires the processor to display the FFT.

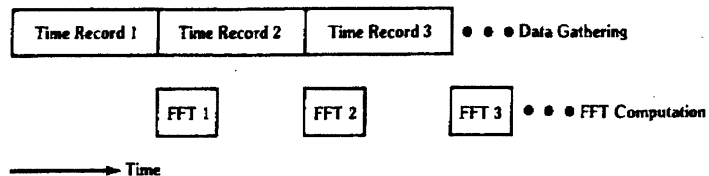


FIGURE 14
NON-OVERLAPPED PROCESSING IS PERFORMED ONLY ON
COMPLETELY NEW DATA (TIME RECORDS) (Ref. 1)

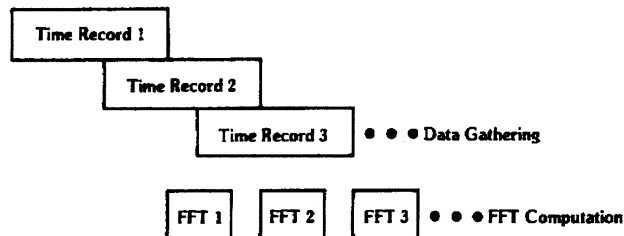


FIGURE 15
OVERLAPPED PROCESSING IS PERFORMED ON DATA THAT COMBINES
OLD & NEW TIME BETWEEN FFT'S REPRESENTING PROCESSING
(Ref. 1)

In essence, a full length time record is first captured. Then portions of new time records are stored and added to portions of the old data. With an overlap of 50%, half of the new time record is taken in and combined with half of the old time record. And, since only half of the new time record is taken in, this cuts the FFT processing time down by a factor of 2. If, on the other hand, one were to ask for 75% overlap, he can cut the time by a factor of approximately 4 times, using only 25% of new information with each calculation. This can greatly speed up the measurement process, especially when capturing 8 or more averages and is acceptable for most condition monitoring measurements - as long as the vibration is periodic (that is, repeating, predictable and without pronounced transients).

Of course, as is the case with many "speed" techniques, one does not get something for nothing. Because an overlapped time record contains old data from before the device adjustment, it is not completely correct. It does indicate the direction and the amount of change, but one must wait for a full time record after the change for the new spectrum to be accurately displayed. Therefore, for measurements such as transient measurements which occur shorter than the time record, overlap processing is useless. For transients longer than the time record, the real-time bandwidth of the analyzer and spectrum recorder is usually a limitation. If it is not, overlap processing allows more spectra to be generated from the transient, usually improving the spectral data captured during the event. In any case, overlap processing can give dramatic reductions in the time required to compute averages and eliminates most of all of the time that would be wasted by the FFT processor waiting for a new time record. For most measurements that would occur during predictive maintenance vibration surveys, overlap processing can be used with a good degree of confidence. Therefore, overlap processing can be quite handy for low-frequency measurements which otherwise would require excessive periods of measurement time.

3.61 EXAMPLE OF SAMPLING TIMES WITH AND WITHOUT OVERLAP PROCESSING

The total time required for these measurements is a function of the time required to calculate the FFT from the measured time waveform as well as the time required for the transducer and data collector system to "settle". It will likely require from 2 to 3 minutes per point on low-speed machines (including settling time). Regardless of the instrument used, the equation which will indicate how long it will take to one time record is as follows:

$$\text{DATA SAMPLING TIME} = \frac{60(\# \text{SAMPLES})(\# \text{AVG.})}{(2.56)(\text{FREQ. SPAN})} = \frac{60(\# \text{FFT LINES})(\# \text{AVG.})}{\text{FREQ. SPAN}} \quad \text{Eqn. 4}$$

where:

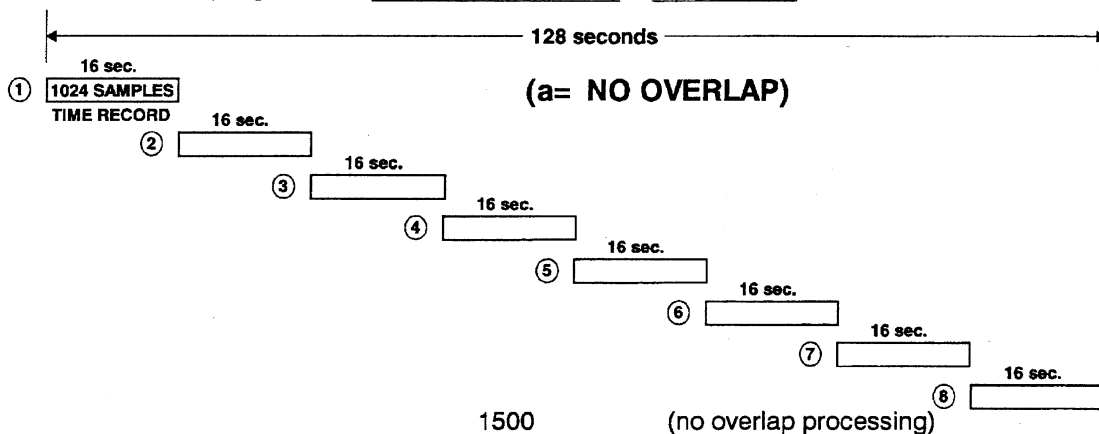
- DATA SAMPLING TIME = Total Sampling Period (setting how often amplitude is measured and stored), seconds. (Assuming no Overlap Processing is involved)
- #SAMPLES = Number of Analog to Digital Conversions to be used to Construct the Time Waveform (1024 samples for 400 line FFT)
- #AVG. = Number of Averages
- FREQ. SPAN = Frequency Span (CPM) (Extending from F_{MIN} to F_{MAX} for systems allowing FFT measurement to begin at a frequency other than 0 CPM).

The following example will illustrate the dramatic reduction in sampling time which can be realized by invoking overlap processing:

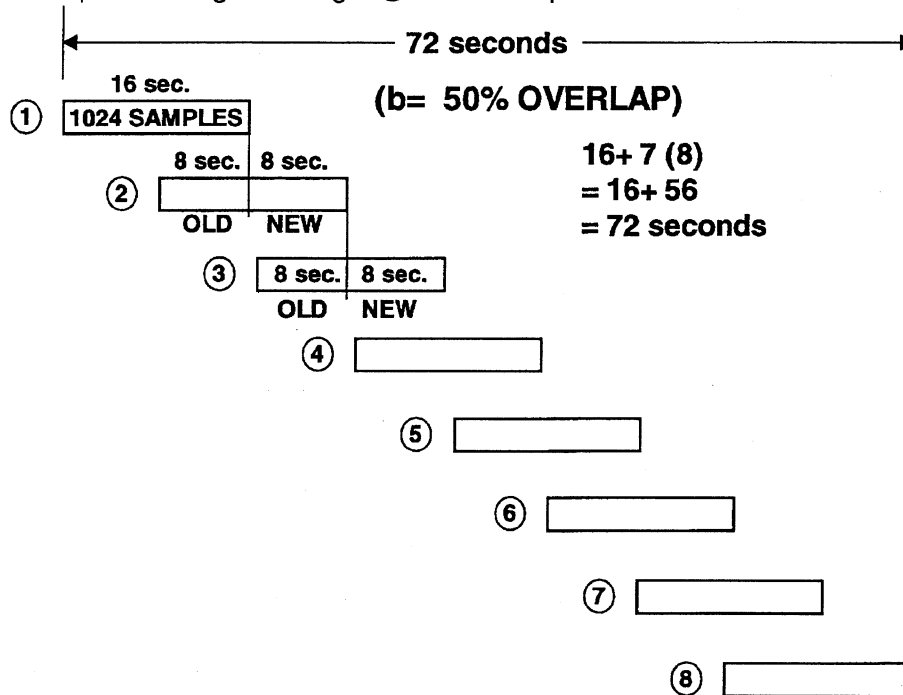
EXAMPLE COMPARISON OF REQUIRED SAMPLING TIME WITH & WITHOUT USING OVERLAP PROCESSING:

For an example, if it was necessary to capture a 400-line FFT, with 8 averages and a frequency range from 0 to 1500 CPM (0 - 25 Hz), it would require 128 seconds just to sample the data (not including transducer and data collector system settling time).

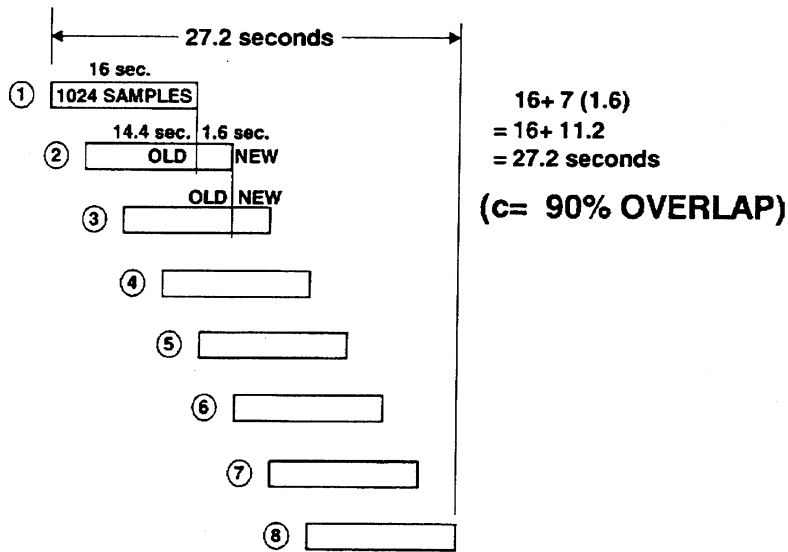
$$\text{Data Sampling Time} = (60)(400 \text{ lines})(8 \text{ ave.}) = 128 \text{ seconds}$$



a) No Overlap Processing - 8 averages @ 16 seconds per time record = 128 sec.



b) With 50% Overlap Processing - 4.5 averages @ 16 seconds for the first time record and 8 seconds for each subsequent time record = 72 sec.



- c) With 90% Overlap Processing - 1.7 averages @ 16 seconds for the first time record and 1.6 seconds for each subsequent time record = 27.2 sec.

While it is not advisable to specify 90% overlapping for condition monitoring surveys (even if the software has the capability), it becomes apparent that much sampling time is saved by specifying at least 50% overlapping. 90% overlap (or above) can be used when the analyst wants to quickly compare spectra at several different points on a machine just to get an idea of relative vibration at various machine measurement locations.

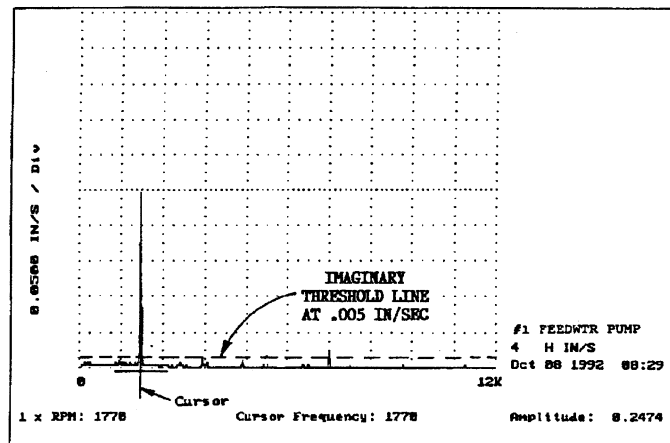
For all practical purposes during PMP surveys, 75% overlapping with only 2 averages need to be specified for F_{MAX} up to 3000 CPM (50 Hz), not 8 averages as used in the above example. Only 4 averages using about 50% overlap need to be specified for frequency ranges with an F_{MAX} between 3000 and 12,000 CPM (50 and 200 Hz); while 8 averages with 50% overlap for frequency ranges from 12,000 to 60,000 CPM (200 to 1000 Hz). For frequency ranges over 60,000 CPM (1000 Hz), 16 averages with no overlapping is recommended.

In the above example if the frequency were increased to 150,000 CPM (2500 Hz), overlapping would have had little value because the sampling time would have dropped from only 1.28 seconds (non-overlapped) down to 0.72 second (50% overlapped). But at 1500 CPM and 2 averages, overlapping does reduce the sampling time considerably. Thus it is a practical technique.

3.7 UNDERSTANDING A VIBRATION SPECTRUM

A spectrum is a graphic display of vibration in the frequency domain. Figure 16 is an example of a spectrum. Note that the X-axis is delineated in CPM with a maximum frequency (F_{MAX}) of 12,000 CPM and the Y-axis is delineated in in/sec of vibration with .050 in/sec peak/division (with 10 divisions, this gives a Full Scale Amplitude of .500 in/sec). The horizontal scale also has ten divisions with each frequency division being one-tenth of 12,000 CPM (or 1200 CPM/division). The addition of a cursor makes determining the frequency and amplitude of an individual peak easy. In Figure 16, the cursor is at the dominant peak. The cursor frequency is indicated as 1770 CPM (the approximate rotating speed of the machine) and its amplitude is .2474 in/sec. The dashed line across the bottom of the spectrum is an imaginary amplitude threshold level of 1%. That is, 1% of the full amplitude scale which is .500 in/sec which amounts to .005 in/sec. Below the spectrum is a listing of the amplitudes, the order (order = multiple of running speed), and the frequencies of all peaks that have an amplitude over .005 in/sec threshold.

It is very important that both the amplitude and frequency values for all peaks in the spectrum be as precise as possible. Amplitude is important, of course, because it gives an idea of the severity of the problem. Frequency, on the other hand, is even more important since it is used to determine the source of the vibration. For instance, In the spectrum in Figure 16 the vibration with the highest peak (at 1770 CPM) equals the rotational speed of the machine (1X RPM). Therefore, the component's rotating speed at 1770 RPM is the source of the dominant vibration. By knowing that the problem source is at 1X RPM, all possible problem sources which cause vibration at 1X RPM can be itemized. At the same time, the possibility of other sources being the problem, such as rolling element bearing, blade pass, or most electrical problems, are eliminated.



1 1/2 x 0.500 = 0.0050		1 x RPM:	1770.00	
Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude		
180	0.102	0.0054		
240	0.136	0.0057		
300	0.169	0.0070		
1050	0.593	0.0062		
1170	0.661	0.0088		
1260	0.712	0.0053		
1350	0.763	0.0053		
1470	0.831	0.0053		
1620	0.915	0.0051		
1770	1.000	0.2474		
2910	1.644	0.0078		
3540	2.000	0.0144		
3660	2.068	0.0055		
4680	2.644	0.0108		
7200	4.068	0.0234		

T - Threshold adjust Q - Quit

FIGURE 16
SPECTRUM WITH CURSOR FREQUENCY AND IDENTIFICATION OF PEAKS
ABOVE A USER-DEFINED THRESHOLD

3.71 EFFECT OF THE NUMBER OF FFT LINES USED ON FREQUENCY ACCURACY

Probably the greatest requirement in determining potential machine problem source(s) is knowing the actual value of the frequency. The determination of an accurate vibration frequency can become clouded or enhanced, depending on how many lines of spectral resolution are used in collecting and displaying the data.

Most computer monitors are able to display 400 lines of spectral resolution (and most data collectors today are set up to capture 400-line spectra on PMP surveys). As discussed previously, the spectral display is divided up by 400 individual lines (or "bins") located adjacent to one another along the frequency axis. Not all bins contain information since the vibration at some frequency bin locations is zero. But the bins that do contain information display an amplitude peak(s) which is proportional to the amount of vibration at that specific bin location or frequency within it. The frequency range per bin (known as frequency resolution) is determined by Equation 5 below:

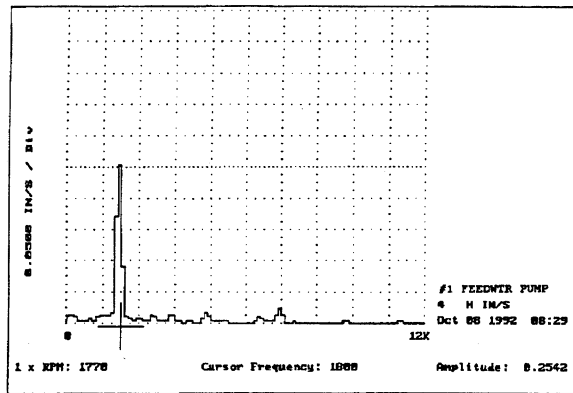
$$\text{FREQUENCY RESOLUTION} = \frac{\text{FREQ. SPAN}}{\text{\# FFT LINES}} \quad \text{Eqn. 5}$$

Where:

$$\begin{aligned} \text{FREQUENCY RESOLUTION} &= \text{DISPLAY RESOLUTION PER LINE} \\ &\quad (\text{CPM/LINE OR HZ/LINE}) \\ \text{FREQUENCY SPAN} &= \text{Frequency Span from } F_{\text{MIN}} \text{ to } F_{\text{MAX}}; \text{ which is often } 0 - F_{\text{MAX}} \\ &\quad (\text{CPM or Hz}) \end{aligned}$$

For example, in Figure 16, the F_{MAX} is 12K CPM (12,000 CPM). If it is divided by 400 lines of resolution, each bin has a 30 CPM range. The 59th bin actually contains vibration information in the range from 1755 to 1785 CPM. The 1770 CPM frequency is in this bin. But it is unknown whether there are one or more peaks within this bin. The vibration peak displayed is the resultant of all frequencies in this bin and the amplitudes of those frequencies within the bin all sum to .2474 in/sec. However, there may actually be only one frequency in this bin after all.

The higher the number of lines of resolution that are specified for a spectrum, the higher will be the precision of the frequency read by the analyzer and displayed. It is important to note that the precision (accuracy range) of any displayed frequency will be the frequency reading plus or minus one-half the resolution. Thus, if a peak at 1800 CPM was displayed in a 12,000 CPM spectrum with a 30 CPM frequency resolution, the accuracy range would be 1800 CPM (± 15 CPM), or from 1785 to 1815 CPM. On the other hand, if a 120,000 CPM spectrum using 400 lines was taken at the same location and showed the same 1800 CPM peak, the accuracy would be $\pm$ half the resolution (300 CPM), or from 1650 to 1950 CPM. Figures 17 thru 20 are spectra with 100, 200, 400, and 3200 lines of resolution. Note that the peaks in the 100-line spectrum of Figure 17 have a very "blocked" appearance with that dominant peak shown at "1800 CPM". With 100 lines of resolution, each bin has a frequency resolution (sometimes called " Δf ") of 120 CPM per line of resolution. Therefore, the frequency value from a 100 line spectrum is considerably inaccurate. Figure 18 has 200 lines of resolution. As a result, its resolution of 60 CPM/line is slightly better than that of Figure 17. Here, the dominant frequency is observed to be at "1740 CPM". Figure 19 is a 400-line spectrum with 30 CPM/line of resolution. The peak is identified as "1770 CPM", which is more accurate than the values in Figures 17 and 18.



001% x 0.500 = 0.0050 1 x RPM: 1770.00

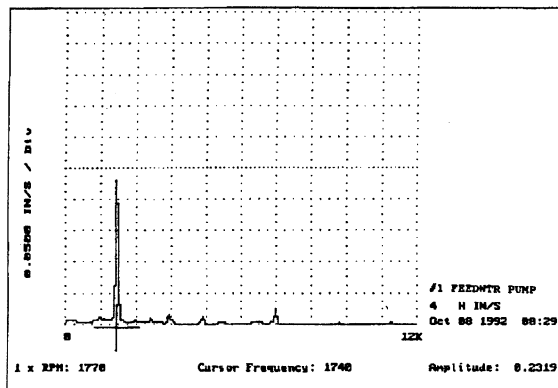
Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29

Frequency	1 x RPM	Amplitude
240	0.136	0.0129
840	0.475	0.0053
1200	0.678	0.0132
1800	1.017	0.2542
2400	1.356	0.0061
2880	1.627	0.0115
3600	2.034	0.0134
4680	2.644	0.0154
6480	3.661	0.0081
7200	4.068	0.0227

FREQ. RESOLUTION = 12,000 CPM = 120 CPM/LINE
100 LINES

T - Threshold adjust Q - Quit

FIGURE 17
12,000 CPM SPECTRUM WITH 100 LINES OF RESOLUTION



1% x 0.500 = 0.0050 1 x RPM: 1770.00

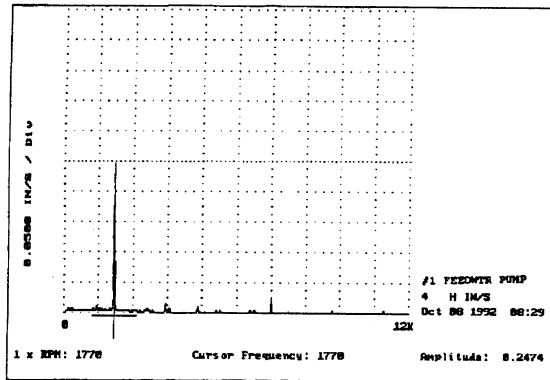
Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29

Frequency	1 x RPM	Amplitude
120	0.068	0.0076
300	0.169	0.0072
1140	0.644	0.0111
1320	0.746	0.0087
1500	0.847	0.0075
1740	0.983	0.2319
2400	1.356	0.0051
2880	1.627	0.0083
3540	2.000	0.0160
4680	2.644	0.0116
7200	4.068	0.0237

FREQ. RESOLUTION = 12,000 CPM = 60 CPM/LINE
200 LINES

T - Threshold adjust Q - Quit

FIGURE 18
12,000 CPM SPECTRUM WITH 200 LINES OF RESOLUTION



1 1/2 x 0.500 = 0.0050 1 x RPM: 1770.00

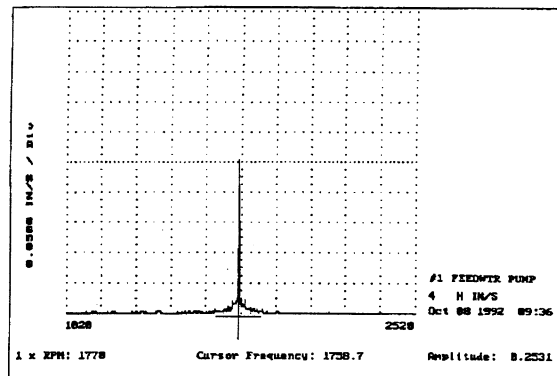
Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29

Frequency	1 x RPM	Amplitude
180	0.102	0.0054
240	0.136	0.0057
300	0.169	0.0070
1050	0.593	0.0062
1170	0.661	0.0088
1260	0.712	0.0053
1350	0.763	0.0053
1470	0.831	0.0053
1620	0.915	0.0051
1770	1.000	0.2474
2910	1.644	0.0078
3540	2.000	0.0144
3660	2.068	0.0055
4680	2.644	0.0108
7200	4.068	0.0234

FREQ. RESOLUTION = 12,000 CPM = 30 CPM/LINE
400 LINES

T - Threshold adjust Q - Quit

FIGURE 19
12,000 CPM SPECTRUM WITH 400 LINES OF RESOLUTION



006 1/2 x 0.500 = 0.0300 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 09:36

Frequency	1 x RPM	Amplitude
1758.7	0.994	0.2531

FREQ. RESOLUTION = 12,000 CPM = 3.75 CPM/LINE
3200 LINES

T - Threshold adjust Q - Quit

FIGURE 20
ZOOM SPECTRA TAKEN FROM A 12,000 CPM FREQUENCY SPAN
SPECTRUM HAVING 3200 LINES OF RESOLUTION

Further accuracy can be achieved by using higher numbers of lines of resolution. Figure 20 is a portion of a 3200-line spectrum. The resolution is 3.75 CPM/line of resolution for the 12,000 CPM frequency span. The dominant vibration is now shown as "1758.7 CPM". The accuracy (precision limits) of this dominant peak would be ± 1.875 CPM (or, again, one-half the frequency resolution, Δf). Thus, the true frequency would actually be between 1756.8 and 1760.6 CPM (29.281 - 29.344 Hz).

So why shouldn't a 3200-line spectra be taken all the time and avoid the hassle of inaccurate data? First of all, a 3200-line spectrum takes 8 times longer to collect than does a 400-line spectrum (as seen by Equation 6 below) which really slows down the data collection process. Secondly, it requires 8 times the storage for a 3200 lines spectrum than for a 400-line spectrum. Judgement has to be exercised since the data collector's spectral storage capacity will be significantly reduced if such high resolution spectra are collected at all points. This will be discussed in more detail in a later section.

When would it be advantageous to take 3200-line spectra? To even detect certain electrical problems (such as cracked/broken rotor bars or end rings) will require high resolution spectra in order to even detect these problems as will be discussed in Chapter 6.12. It is also extremely useful when vibration frequencies from different machines are very close to one other. In other instances, some bearing frequencies and harmonics are commonly close to exact multiples of the operating speed. In another case, a pump may have an impeller with 6 vanes and rolling element bearings which have a defect frequency of 6.03X RPM. In this case, it may require a high resolution spectrum to separate the bearing frequency (6.03X RPM) from the vane passing frequency (6X RPM). Here, a high degree of accuracy is needed. However, for general data collection, 400 lines of resolution produces adequate accuracy.

$\text{DATA SAMPLING TIME} = \frac{60(\# \text{ FFT LINES})(\# \text{ AVG.})}{F_{\text{MAX}}}$	Eqn. 6
--	--------

Equation (6) is a modification of Equation (4).

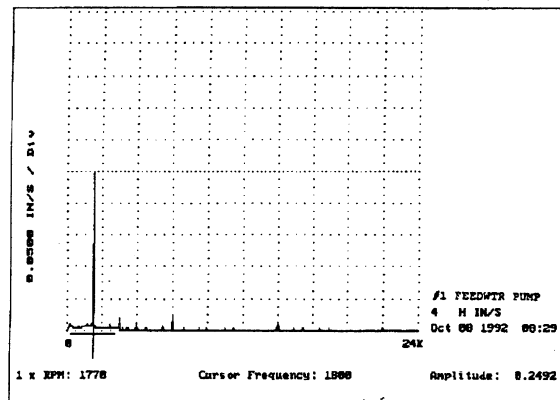
Where:

DATA SAMPLING TIME	= Time to calculate FFT from Time Waveform (sec); Assuming no Overlap Processing.
# FFT LINES	= Number of FFT Lines or Bins in Spectrum
# AVG.	= Number of Averages
F_{MAX}	= Frequency Span (CPM)

3.72 EFFECT OF THE FREQUENCY SPAN USED ON FREQUENCY ACCURACY

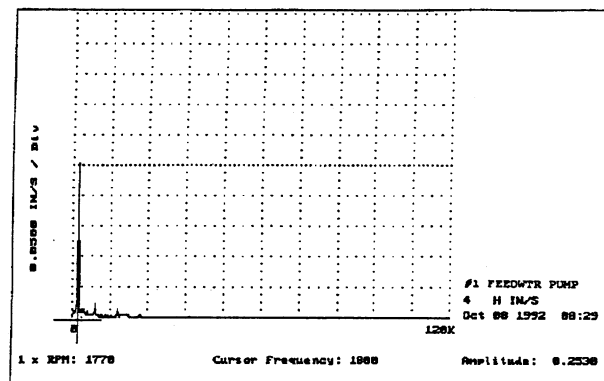
Another factor affecting resolution is the frequency range (F_{MAX}) setting. The larger the frequency range, the less accurate the frequency readings will be and the wider each line of resolution (bin) will be. Figure 21 is a 400-line spectrum with a 24,000 CPM F_{MAX} . The resolution of this spectrum is 24,000/400 lines of resolution = 60 CPM/line of resolution. In comparison with the 12,000 CPM F_{MAX} , 400-line spectrum of Figure 19, its frequency definition is only one-half as accurate. Figure 22 is a 120,000 CPM F_{MAX} spectrum with 400 lines of resolution. Here, the resolution is 300 CPM/line of resolution which means that the accuracy of any displayed frequency will be ± 150 CPM (or plus or minus one-half the resolution). In this case, its frequency definition is only 1/10 of the precision in Figure 19. In some cases, this amount of accuracy may be all that is needed.

Trade-offs must be made to retain accuracy. The analyst needs to specify an F_{MAX} large enough to include all the vibration data needed, while not (at the same time) unduly sacrificing the accuracy in the process. Technical Associates has a proven method for choosing F_{MAX} values for data collectors. This method will be discussed in detail in a later chapter.



1 1/2 x 0.500 = 0.0050		1 x RPM:	1770.00	
Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude	FREQ. RESOLUTION = <u>24,000 CPM</u> = 60 CPM/LINE 400 LINES	
180	0.102	0.0095		
780	0.441	0.0050		
1020	0.576	0.0066		
1140	0.644	0.0076		
1380	0.780	0.0097		
1500	0.847	0.0078		
1800	1.017	0.2492		
2940	1.661	0.0083		
3600	2.034	0.0178		
4740	2.678	0.0105		
7200	4.068	0.0253		
14400	8.136	0.0111		

FIGURE 21
24,000 CPM SPECTRUM WITH 400 LINES OF RESOLUTION



1 1/2 x 0.500 = 0.0050		1 x RPM: 1770.00		
Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 08:29
Frequency	1 x RPM	Amplitude	FREQ. RESOLUTION = <u>120,000 CPM</u> = 300 CPM/LINE 400 LINES	
300	0.17	0.0098		
1800	1.02	0.2530		
3000	1.69	0.0117		
3600	2.03	0.0137		
4800	2.71	0.0091		
7200	4.07	0.0230		
14400	8.14	0.0111		

FIGURE 22
120,000 CPM SPECTRUM WITH 400 LINES OF RESOLUTION

3.73 IMPROVING THE FREQUENCY RESOLUTION WITH "ZOOM" - BAND SELECTABLE FOURIER ANALYSIS

Recall that a Fast Fourier Transform is equivalent to a set of filters, starting at zero Hertz, equally spaced up to some maximum frequency. Therefore, the frequency resolution is limited to the maximum frequency divided by the number of "filters".

In addition to the examples given in Section 3.71, suppose we needed to measure a small signal which is very close in frequency to a large one. For example, the condition of an electric motor can be assessed by acquiring motor current spectra using a current transformer (CT) as the transducer. When evaluating motor current spectra for rotor problems, one will have to compare the amplitude at electrical line frequency (F_L) with that at what is known as the first pole pass frequency (F_p) to its left (where pole pass frequency = slip frequency X #poles as will be discussed in Chapter 6.12). In this country, the line frequency (F_L) is normally 60 Hz (3600 CPM) and will dominate motor current spectra, thereby requiring the analyst to use "log amplitude" to even see the pole pass frequency sidebands surrounding it. And, typically these pole pass frequencies will only range from about 30 CPM (0.5 Hz) to 180 CPM (3.0 Hz). Therefore, precise frequency resolution is required for this measurement. One obvious advantage in decreasing the frequency resolution is the ability to distinguish closely spaced peaks in the spectrum.

Two ways of improving the frequency resolution were discussed in the last section. One way to improve frequency resolution is to increase the number of FFT lines in the spectrum and then display only those lines of interest. Another way is "frequency zoom". Frequency zoom is the ability to concentrate all 400 lines of resolution in the frequency band of interest. Figure 23 is a display of the example above. Figure A is the measurement without using zoom. It is a partial display (40 lines of a 400 line spectrum that spans from 0 to 200 Hz or 0 - 12,000 CPM). Figure B is the same measurement using zoom. With zoom, all 400 lines of the spectrum are concentrated between 50 and 70 Hz (3000-4200 CPM given a 3.0 CPM resolution). In Figure A, the sidebands are barely above the noise floor - if they were much lower they would not be measurable. The noise floor in Figure 23B is reduced, making the sidebands much more visible.

There is a difference between "zoom" and "true zoom". Some analyzers can compute spectra with up to 12,800 lines of interest. If only a portion of the 12,800 lines are displayed, the result is the same as zoom measurement. Although some vendors may call this zoom, it is actually "display expansion". It is not "true zoom". True zoom is implemented by selecting the frequency span of interest before performing the FFT, not afterwards. Thus, a 400 line FFT would be adequate for most "true zoom" measurements because all 400 lines of resolution are concentrated between the frequencies of interest (F_{MIN} to F_{MAX}).

But the frequency resolution would not be improved by selecting a so-called "zoom" from a spectrum stored with only 400 FFT lines. By selecting a scaling factor of 4 on the analyzer only expands the frequencies by that factor proportionally. As shown in Figure 24, F_{MAX} was made to equal 15,000 CPM rather than the original 60,000 CPM. The frequency expanded spectrum allows for a better look at the shape of each frequency peak, but it does not improve the frequency resolution.

As was mentioned previously, one way to improve resolution is to increase the number of lines in the FFT and then display them. For instance, capturing and storing a 3200-line FFT does improve frequency resolution by a factor of 8X (3200/400) as shown in Figure 25. However, a 3200-line FFT also requires 8X the measurement time in the field and requires 8X the storage space of a 400 line FFT back in the computer.

Figure A - Spectrum of motor current, measured without zoom

DISPLAY IS ONLY 40 LINES OF A 400-LINE FFT

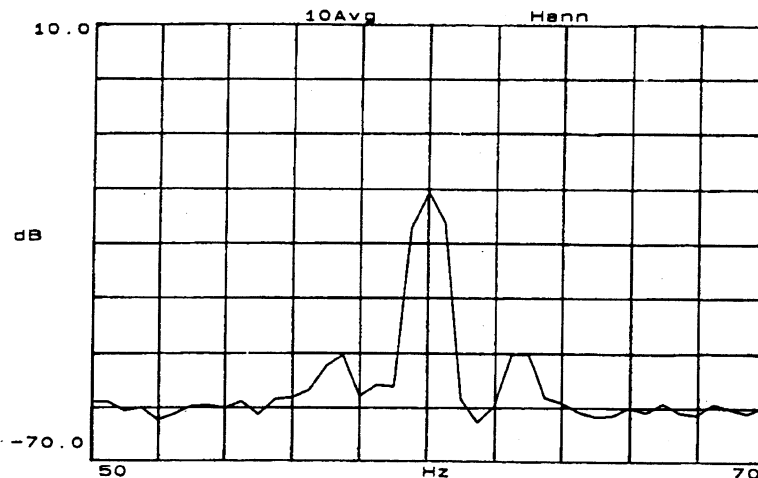


Figure B - Spectrum of motor current, measured with zoom

**DISPLAY IS 400 LINES CONCENTRATED (AROUND 60 Hz)
BETWEEN 50 TO 70 Hz**

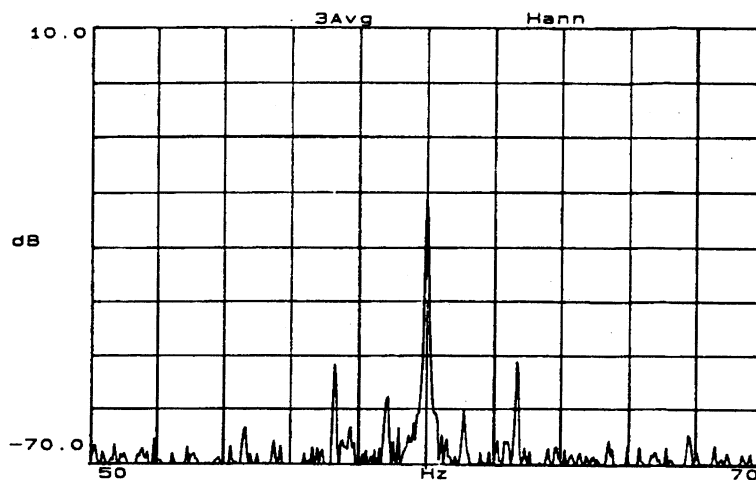


FIGURE 23
TWO EXAMPLES OF MOTOR CURRENT SPECTRA, WITH AND
WITHOUT "ZOOM" (Ref. 5)

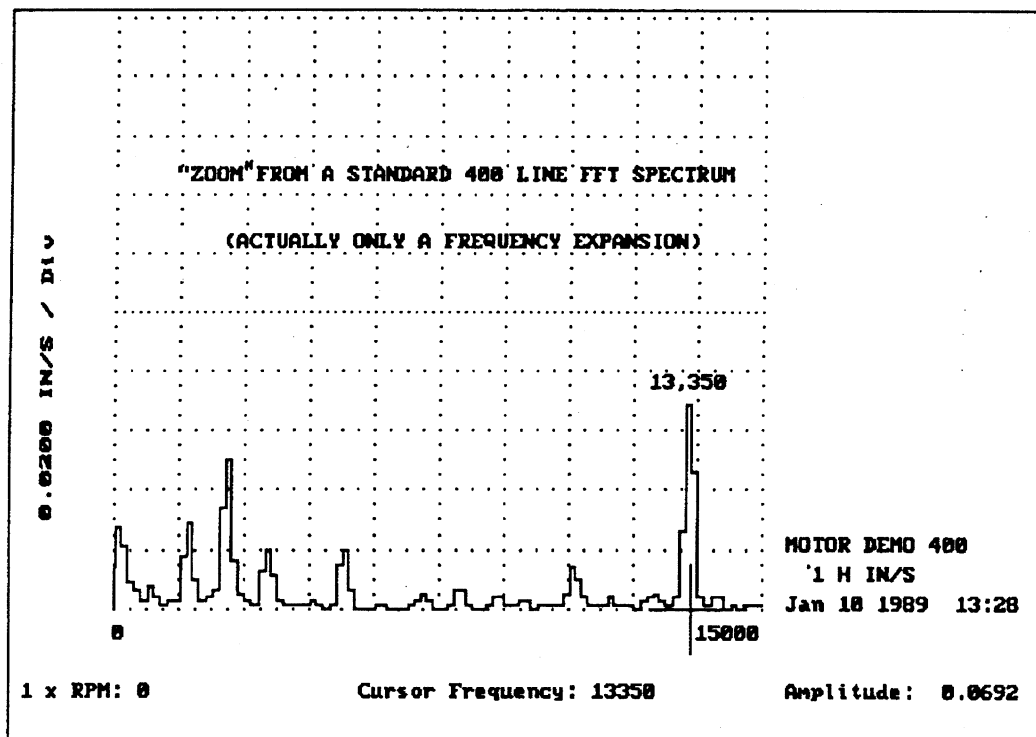
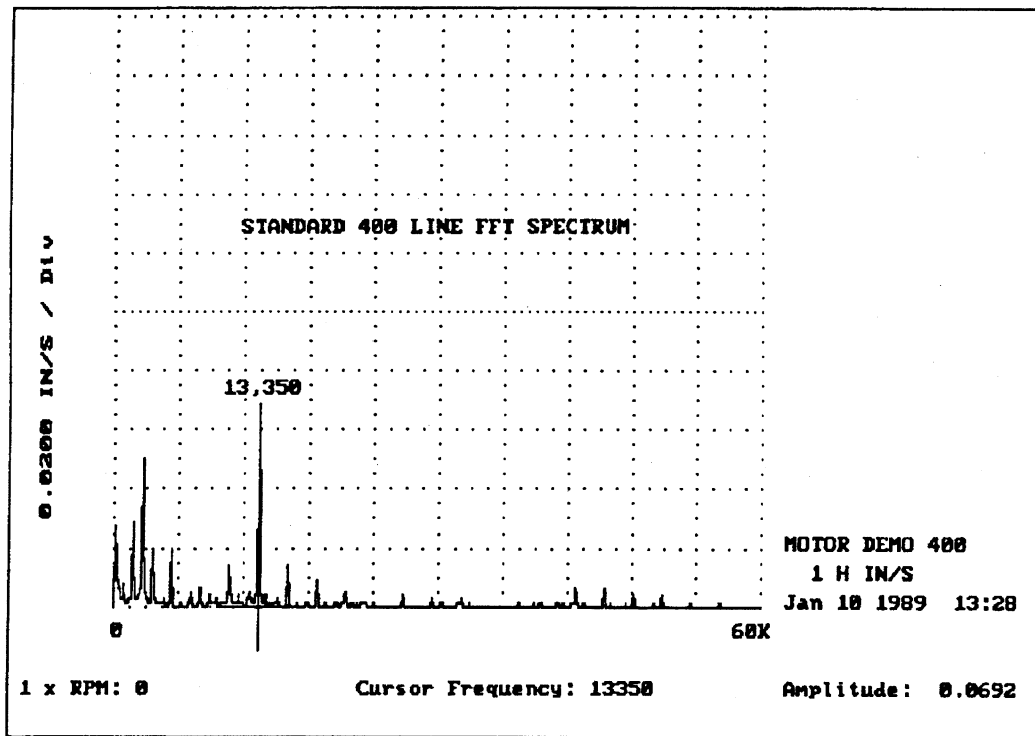


FIGURE 24
EXAMPLE OF A SO-CALLED "ZOOM" SPECTRUM

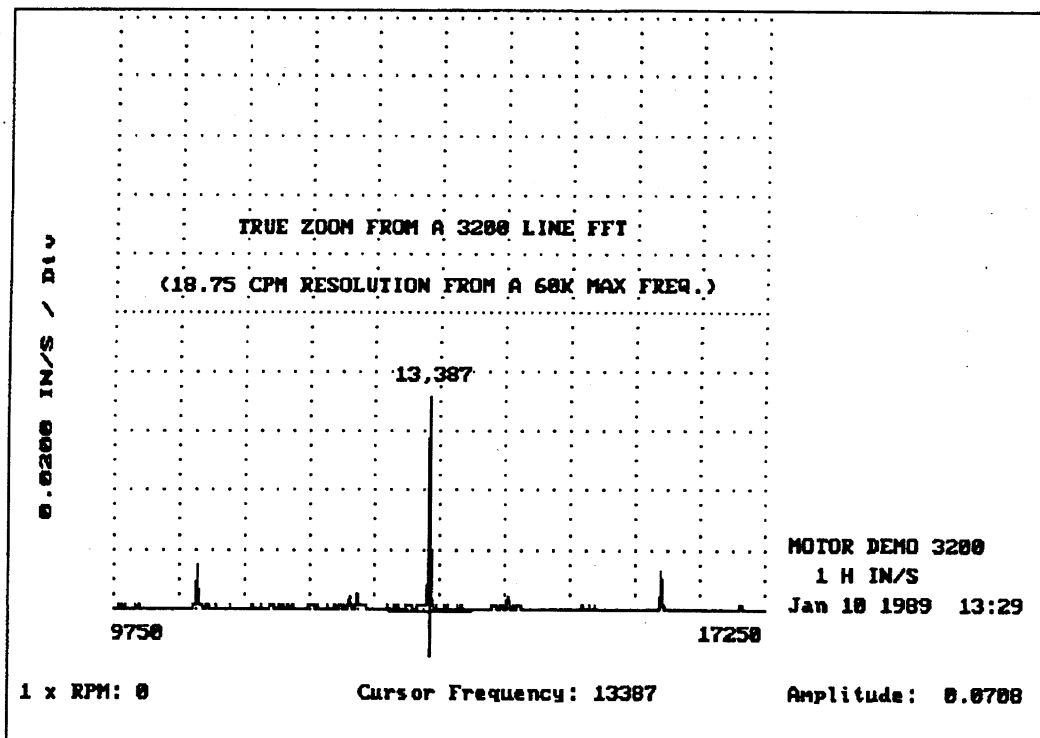
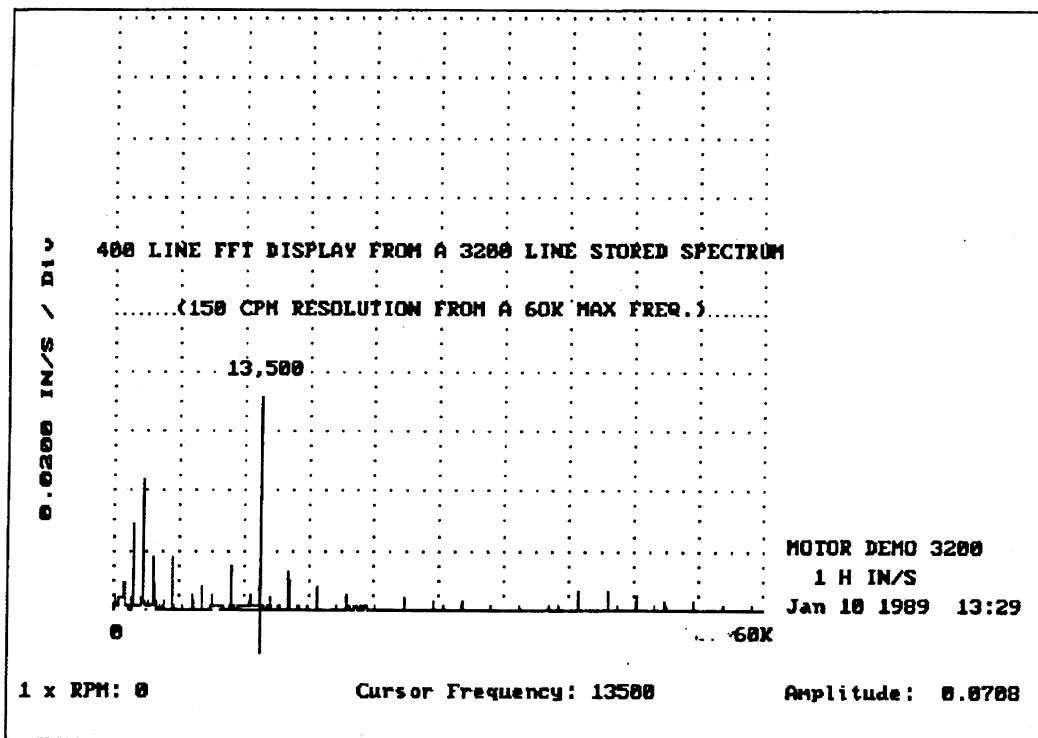


FIGURE 25
EXAMPLE OF A "TRUE ZOOM" SPECTRUM (Ref. 5)

Also mentioned previously, was selecting the frequency band of interest before performing the FFT. This is practical if the frequency span of interest is known before starting the measurement, provided the analyzer is capable of performing this band selectable Fourier analysis. The advantages are apparent. First, "true zoom" can reduce the measurement time considerably and reduce data storage because a 400 line FFT only takes 1/8 the memory, compared to 3200 lines. However, in the real-world, the frequency span which needs to be zoomed is rarely known, unless it was derived from previous data. In this case, it is better to use display expansion, a pseudo "true zoom", because the improved frequency resolution will accurately identify specific peaks of interest as shown in Figure 25. Here, the 13,500 CPM peak seen in the upper 400-line display, is more accurately identified as a 13,387 CPM peak in the "zoom" of the lower display using 3200 lines. Also, the sidebands can be readily identified.

The point to remember is merely performing a display expansion on a "zoom" of a 400-line FFT will not approve the frequency resolution. Only by increasing the lines of resolution and/or decreasing the F_{MAX} will the "zoom", "display expansion" or "true zoom" actually improve the frequency resolution so that spectral peaks can be seen and accurately identified.

3.74 IMPROVING THE PRECISION OF THE SPECTRUM BY FREQUENCY & AMPLITUDE INTERPOLATION

Only the latest vibration instruments and software are capable of displaying as much as 12,800 line spectra. A technique which can be employed by an analyst to enhance the frequency and amplitude values from common 400 line spectral data will be discussed in this section. A set of equations will be provided to improve frequency and amplitude readings by as much as 100 times, given any frequency span and the number of FFT lines of resolution. Of course, for a smaller frequency span with a greater number of FFT lines, the calculated values will be closer to the actual value.

$$f_i = f_p + \Delta f_c \quad (\text{if } f_s > f_p) \quad \text{Eqn. 7A}$$

$$f_i = f_p - \Delta f_c \quad (\text{if } f_s < f_p) \quad \text{Eqn. 7B}$$

3.741 Equations for Closest Frequency Interpolation (f_i)

where:

- f_i = Interpolated Frequency calculated to be closest to the True (actual) Frequency, dependent on Frequency Resolution, Δf (see Figure 26 (A-2))
- f_p = Displayed Frequency on Spectrum (Closest Frequency Peak to f_i , dependent on Frequency Resolution used, Δf)
- Δf_c = Frequency Correction, or distance from f_p to f_i (CPM or Hz)
- f_s = Frequency (either above or below f_p) in the next bin (f_s) (see Figure 26 (A-1))

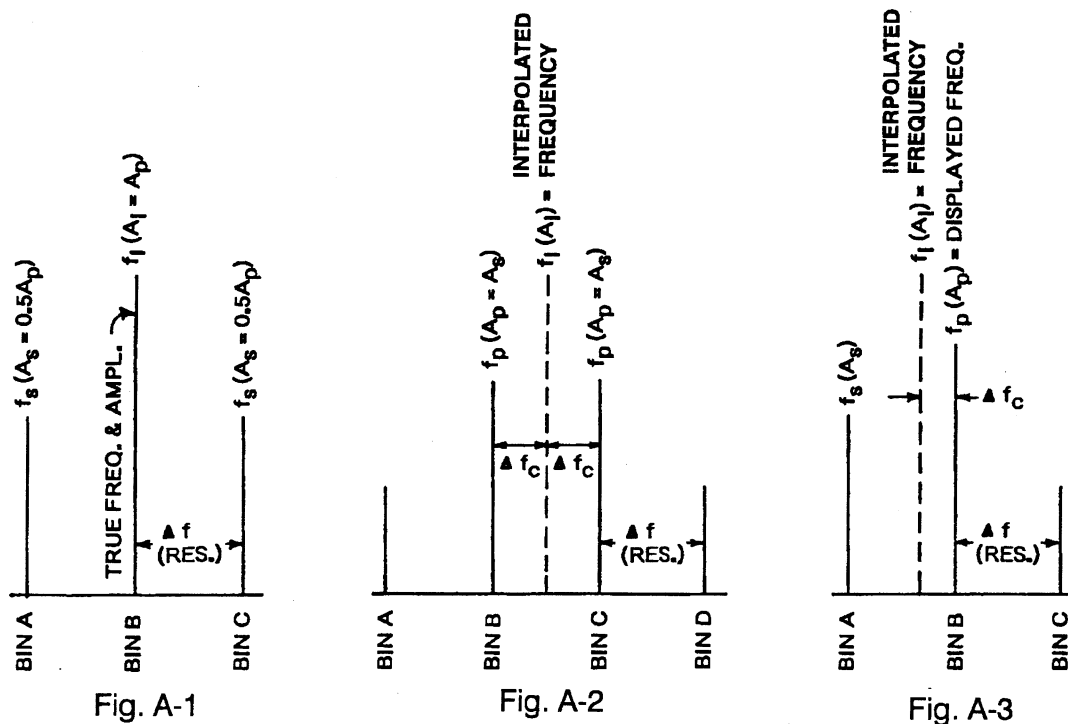


FIGURE 26
IMPROVEMENT OF DISPLAYED AMPLITUDE & FREQUENCY READINGS BY
BIN DISTRIBUTION CALCULATIONS (SEE EQUATIONS 7A THRU 9)

$$\Delta f_c = \frac{2A_s - A_p}{A_s + A_p} \Delta f$$

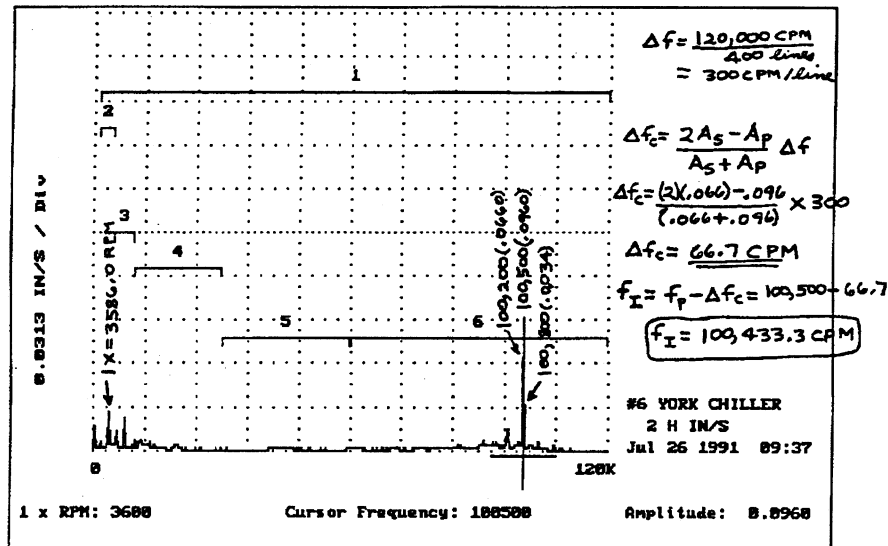
Eqn. 8

The Frequency Correction (Δf_c) can be calculated as follows:

where:

- A_s = Amplitude of the Frequency in Bin f_s (see Figure 26 (A3))
- A_p = Amplitude of the Frequency in Bin f_p (see Figure 26 (A3))
- Δf = Frequency Resolution = FREQ. SPAN/#FFT LINES (Hz or CPM/Line)

Therefore, by using Equations 7 and 8, the precision of the displayed frequencies can be improved very much by examining closely the amplitudes of the bins to the right and left of each frequency of interest one at a time, choosing the higher amplitude of the two bins adjacent to it. Examples that follow will demonstrate the use of this technique, such as in Figures 27 and 28.



003% x 0.200 = 0.0060 1 x RPM: 3600.00

Machine #6 YORK CHILLER		Pos 2	Dir H	Units IN/S	Date Jul 26 1991 09:37
Frequency	1 x RPM	Amplitude	MEASURED VALUES		CALCULATED VALUES
300	0.08	0.0170	3300 (.0150) = A_{s1}	USE HIGHEST AMPLITUDES	$f_I = 3586.0 \text{ RPM}$
3600	1.00	0.0280	3900 (.0130) = A_{s2}		
5400	1.50	0.0140	6400 (.0130) = A_{s2}		
7200	2.00	0.0240	7500 (.0110)		
9600	2.67	0.0070			
11100	3.08	0.0080			
13200	3.67	0.0060			
91200	25.33	0.0070			
93300	25.92	0.0060			
96600	26.83	0.0160	96,300 (.0090) = A_{s1}	CORRECTED ORDER# (1X SIDEBAND) →	$f_I = 96,735.5 = 26.98 \times \text{RPM}$
99300	27.58	0.0060	96,900 (.0150) = A_{s2}		
100500	27.92	0.0960	100,200 (.066) = A_{s2}		
104100	28.92	0.0060	100,500 (.034)		
				CORRECTED ORDER# = 28.00 X RPM	

CONCLUSIONS:

(POSSIBLE VANE PASSES)
NOT LIKELY A BEARING

T - Threshold adjust Q - Quit

ALSO, IF THIS IS VANE PASS,
A NARROW BAND OF HIGHER
AMPLITUDE NEEDS TO BE
BUILT AROUND IT (±1X RPM)
RECOMM. ALARM = .150 in/sec
(if truly Vane Pass)

FIGURE 27
EXAMPLE OF USING EQUATIONS FOR FREQUENCY INTERPOLATION
FROM MEASURED FREQUENCY AND AMPLITUDE VALUES

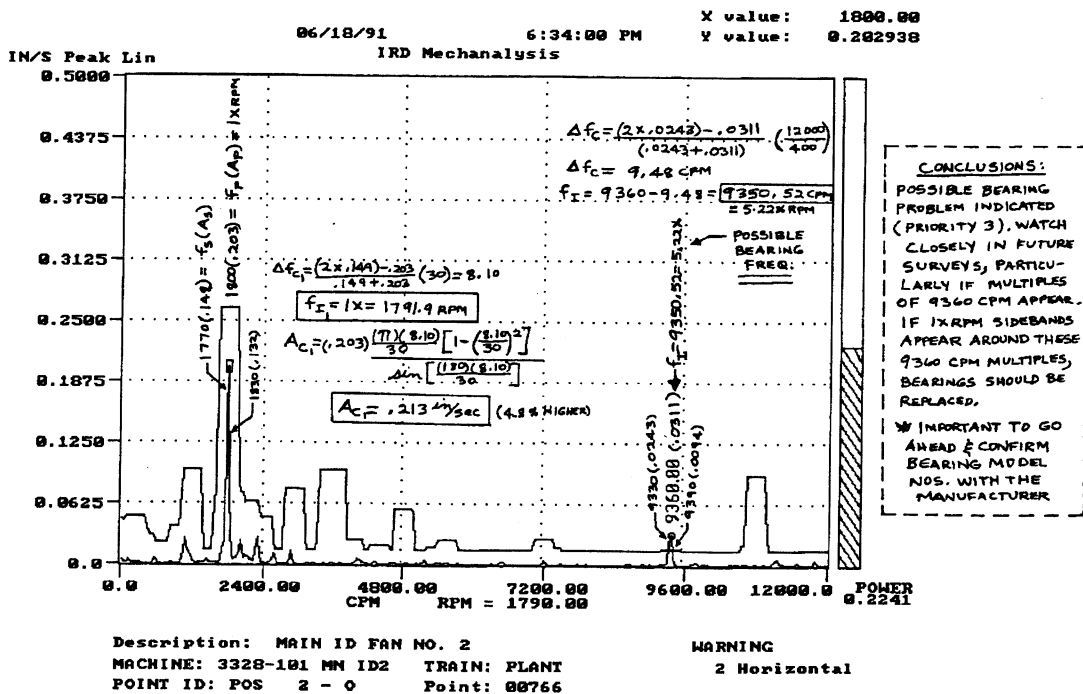


FIGURE 28
EXAMPLE OF USING EQUATION FOR AMPLITUDE INTERPOLATION FROM
MEASURED FREQUENCY AND AMPLITUDE VALUES

3.742 Equation for Closest Amplitude Interpolation (A_I)

$$A_I = A_p \frac{\frac{\pi \Delta f_c}{\Delta f} \left[1 - \left(\frac{\Delta f_c}{\Delta f} \right)^2 \right]}{\sin \left[\frac{180(\Delta f_c)}{\Delta f} \right]} \quad \text{Eqn. 9}$$

Similarly, amplitude readings can be improved by examining the same data shown in Figure 27 as follows:

where:

- A_I = Interpolated Amplitude calculated to be closest to the True Amplitude of the True Frequency, dependent on the Frequency Resolution used.
- A_p = Amplitude of Closest Frequency Bin (f_p) to the True Frequency, (where units of amplitude can be g, in/sec, mils, dB, etc.)

3.75 IMPROVING THE AMPLITUDE ACCURACY BY CHECKING THE BANDWIDTH

The previous technique will have limitations any time that two frequencies exist within one bandwidth. For example, in the case where a 2-pole motor has a running speed of 3590 RPM: if a frequency span of 60,000 CPM and 400 lines of resolution is employed, the 48th line (or bin) will contain both the 2X RPM (7180 CPM) and the 2X electrical line frequencies (7200 CPM). This will adversely affect the interpolated calculation of the amplitude of the peak and the value of frequencies (in fact, the frequency interpolation technique will not work if two or more frequencies are either within one bin or in two bins adjacent to one another).

Recall from Sections 3.41 and 3.42 that the mathematics behind the Fourier Transform assumes that the input data is periodic. (That is, an exact integer number of periods of input data will be supplied). Rarely is this the case. Instead, there is generally non-periodic input data that is supplied to the transform. Without windowing, the FFT will find the discontinuity, between the end of one sample (data buffer) and the beginning of the next, and calculate the extra frequency components that appear as sidebands about the frequency of interest. Recall that the calculation of these extra frequencies is known as "spectral leakage".

These Sections also showed that the spectral leakage can be reduced by multiplying the time sample (the frame of time domain data going to the FFT algorithm) by a window, shaped to smooth the transition between the end of one time sample and the beginning of the next. Recall from Section 3.43 that the most commonly used window is the Hanning Window (it is also known as the cosine squared window; it has zero magnitude and slope at both ends, and a magnitude of 1 in the middle). That is, the FFT is forced to concentrate on the center portion of the sampled data. But even with the Hanning Window, some leakage will be present when an FFT is performed on a non-integer number of cycles of a signal. What this implies is that the frequency component of a signal does not exactly correspond to one of the lines (bins) in the frequency spectrum.

For example, the Hanning window can spread a single sine wave across three spectral lines, whereas a single sine wave would appear as a single spectral line if no window is used. Even though windowing reduces spectral leakage, there is still some spreading and reduced amplitude when any frequency does not coincide exactly with the center of an FFT line (or bin).

This phenomenon presents a problem. If a single sine wave can be spread across three (or four) lines, how can one realistically determine what the narrowest bandwidth can be so as to be able to separately display two frequencies (closely spaced to one another), each showing its peak amplitude?

The following calculation is a further refinement of what has been presented so far. At this point, we are familiar with the definition of frequency resolution as being the frequency span (used) divided by the number of FFT lines. What needs to be done next is to modify the frequency resolution or "bandwidth" by introducing a window factor as follows:

$$\text{BANDWIDTH} = \frac{(\text{FREQUENCY SPAN})(\text{WINDOW NOISE FACTOR})}{\# \text{ FFT LINES}}$$

Eqn. 10

Where:

BANDWIDTH = The spacing between frequencies at which a bandpass filter attenuates the signal by 3 dB (down to 70.7% of its flat response). Two individual frequencies must have at least two bandwidths separation for them both to be displayed on an FFT spectrum with each showing its actual "peak" amplitude.

WINDOW NOISE FACTOR = Effective Noise Bandwidth
 = 1.5 for Hanning Window
 = 1.0 for Uniform Window (also called Rectangular Window)
 = 3.6 - 4.2 for Flat Top Window (depending on analyzer vendor)

If no window (Uniform window) is used, one would use 1.0 for the window factor, whereas if a Hanning window is used, the window factor is 1.5. If other windows are used in an analysis, be aware that the results from Equation 8 (calculation of Δf_c frequency correction) and Equation 9 (calculation of Interpolated Amplitude A_i) will be altered. To simplify this discussion, only the window factor for the Hanning window will be highlighted.

It would appear that by multiplying the resolution by 1.5, the bandwidth would be 1.5 times larger than it would be if the frequency resolution was left as is. Mathematically, that is true. However, do not miss the purpose of this exercise. It is done to determine the narrowest bandwidth that can be used to separately display two frequencies closely spaced to each other. It is meant to compensate for the fact that the signal was spread across three spectral lines, instead of in one line, since a Hanning window was used. Especially when setting up a database, what we are trying to determine is, if we are aware of the presence of closely-spaced frequencies, what would the maximum bandwidth have to be in order to display two distinct frequencies? This would particularly apply in the analysis of AC induction motors. The presence of closely-spaced pole-pass frequency sidebands around 1X RPM or around the 2X, 3X, 4X and 5X RPM harmonic peaks is a strong indication of broken or cracked rotor bars and/or high resistance joints at the shorting ring. Also, in the case of a two-pole motor, 2X RPM will be extremely close to the 7200 CPM (2X F_s) peak. Or, in the case of a four-pole motor, the 4X RPM peak will be extremely close to the 7200 CPM peak.

For most of the frequency ranges chosen, an analyzer will not display two individual frequencies; in these examples, it may display only one frequency in the above cases. In order to see two separate closely-spaced frequencies, it will be necessary either to increase the number of FFT lines, lower the frequency span, or generate a "zoom" spectrum.

This technique is illustrated in Figure 29. Figure 29A is a spectral plot from a 2-pole AC motor driving a centrifugal air compressor. The measurement point is at the inboard bearing of the motor in the horizontal direction. The F_{MAX} was set at 30,000 CPM using 400 lines of resolution in the analyzer. Thus, the resolution was 75 CPM/line. Multiplying this value by 1.5 gives us a bandwidth of 112.5 CPM. Notice that all the peaks are certainly more than 112.5 CPM apart. In this case, it appears that these peaks are running speed harmonics of 3600 CPM (approximately 1X RPM). In fact, this particular motor was originally diagnosed by this client as having a mechanical looseness problem.

However, a further analysis was done. Using a real-time analyzer, the frequency span was now set to 6000 CPM (100 Hz), but the cursor was set on the 1X RPM peak and a zoom was initiated. Figure 29B is a spectral plot of this zoom, which is 600 CPM wide over 400 lines of resolution.

The resolution is 1.5 CPM/line and the bandwidth is 2.25 CPM. In this case, pole-pass sidebands can clearly be seen around the 1X RPM peak, (two "families" of them in fact). In addition, the line frequency ($F_L = 3600$ CPM) was also present. These peaks certainly were not evident in Figure 29A.

Similarly, Figure 29C is a spectral plot of a zoom around the 2X RPM peak. The frequency span was now set to 12,000 CPM (200 Hz), but the zoom displays 1200 CPM (10% of 12,000 CPM) over 400 lines. Therefore, the resolution is 3 CPM/line and the bandwidth is 4.5 CPM. Once again, the pole-pass frequencies can be clearly identified around 2X RPM. These peaks were also not evident in Figure 29A.

Likewise, Figure 29D is a spectral plot of a zoom around the 3X RPM peak. In this case, the resolution is also 3 CPM/line and the bandwidth is 4.5 CPM. Two "families" of pole-pass frequencies can be seen with a third just starting up. Notice that the amplitudes varied from .007 in/sec to .051 in/sec. Does this motor have an electrical problem or not?

It is generally accepted that if pole pass frequencies (F_p) are present around 3 or 4 running speed harmonics with amplitudes above .012 - .015 in/sec, the motor likely has serious problems. But how will we be able to see these pole pass sidebands if we don't consider the bandwidth first?

There may be a number of practical reasons why bandwidth needs to be given consideration. The most outstanding purpose behind calculating the bandwidth is to assure that there will be enough frequency separation to clearly define two or more spectral peaks closely-spaced as was seen with pole-passing frequencies. Another reason, but lesser in importance, is that the amplitude of a sideband in relation to the fundamental frequency is often used as a "gauge" to determine the seriousness of a defect or problem. Thus, it is important that there is enough resolution and bandwidth to be able to display the sideband frequencies so that their amplitudes can be measured. Coupled with zoom analysis, this technique provides the analyst with a means of displaying all necessary spectral peaks.

3.76 EFFECT OF DYNAMIC RANGE ON FREQUENCY AND AMPLITUDE DISPLAY

Apart from specifying the optimum F_{MAX} , number of FFT lines, and checking the bandwidth, there is another problem which could be responsible for not being able to see sidebands and other small peaks in the presence of larger peaks. The problem could be that the instrument used does not have enough "dynamic range".

Dynamic range is defined as "the amplitude ratio (in dB) between the maximum input level and the self-noise (or noise floor) of an instrument or system before distortion." The noise floor is that area along the spectrum where tiny spectral peaks can be seen. If the amplitudes of the peaks of interest are high enough to stand out like "trees in the grass", they are often easy to measure and evaluate. The problem becomes more complex when the spectral peaks are barely distinguishable from the noise floor; it then becomes difficult to see discrete frequencies. This is a primary concern, especially when performing low frequency, low amplitude measurements.

The vibration signal is analogous to a ship on an ocean, where sea level is equivalent to the noise floor of the measurement. The higher the ship rides in the water, the more information about it will be optically available and the easier it is to detect on the horizon - submerged ships go undetected. If a picture is taken of the ship, no amount of photographic enhancement will display the part of the ship which is below the surface or of any portion of submerged ships. The point is, noise can obscure spectral data, alter amplitude information, and render measurements useless. It is important to use an instrument that is equipped to maximize the signal-to-noise ratio of the vibration measurement. Therefore, the dynamic range on an instrument vendor's specification sheet needs to be given more consideration. In order to do this, one has to become familiar with terms, such as "dB", "bit", and "dynamic range".

For instance, one might read that a certain vendor sells an instrument that is an "8-bit data collector with a dynamic range of 48 dB" or another vendor will sell "a 12-bit or 14-bit instrument with a dynamic range of 72 dB and 84 dB", respectively. What does this mean? Does it make any difference?

Recall from Section 3.1 that microprocessors work in powers of 2, called binary numbers. Thus, 1024 intervals or equally-spaced samples equal 2^{10} intervals. If an analog-to-digital instrument was designed to do this, it would be referred to as "a 10-bit A/D instrument".

The chart in Figure 33 helps to demonstrate the difference between an 8-bit and a 12-bit data collector. Eight bit (8-bit) data collectors have some serious shortcomings about which an analyst needs to be aware. The dynamic range of an 8-bit instrument is about 48 dB, whereas the dynamic range for a 12-bit instrument is approximately 72 dB, as shown in the figure. The difference in dynamic range is 24 dB; that is, the 12-bit instrument is 24 dB more sensitive. What does this mean? How much more sensitive does that make the 12-bit instrument than the 8-bit instrument?

By using the formula provided in Figure 33, one could find the antilog of (dB/20), which simply involves dividing 24 dB by 20 which is equal to 1.2 ($10^{1.2}$). Then, using a calculator, hit the "INV" or "2ndF" and the "LOG" key; enter in 1.2, and hit the "=" key. The answer is 15.85 or 16. This means that the 12-bit A/D instruments are 16 times more sensitive than the 8-bit instruments. Being more sensitive means that it is able to much more readily respond to or record small changes or variations. Thus, the more sensitive an instrument is, the more likely it will display small peaks that are sidebands and harmonics of fundamental frequencies. Not only will they be displayed, but the amplitude and frequency values will be more accurate.

To illustrate the difference between an 8-bit and 12-bit instrument, we can examine what happened on a forming fan at a major fiberglass manufacturing client when measurements were simultaneously taken with both an 8-bit instrument (IRD 818) and a 12-bit instrument (IRD 890) on the same bearing housing, with the same IRD 970 accelerometer connected to each instrument (initially connected to the 890; then the 970 was left in place, but the connector at the data collector was removed from the 890 and attached to the 818). Note that the cascade spectrum in Figure 34 shows that measurements had been taken for several surveys with the IRD 818. Then, when Technical Associates visited the plant with an IRD 890, an abruptly different signature was noticed by our analyst as per the 12/3/90 spectrum in Figure 34. This was reported to the plant and a decision was made to conduct the simultaneous measurement test at first opportunity during the seminar.

On 12/5/90, this simultaneous measurement test was performed. First the 12-bit IRD 890 captured the data at 13:51; then the transducer cable connector was removed from the 890 and connected to the 818. The 818 captured the spectrum at 13:52 which is also shown in Figure 34. The important difference between the spectra captured by the 8-bit and 12-bit instrument is the presence of several families of 1X RPM sidebands around the inner race defect frequency (BPF1) located at 8.18X RPM in the IRD 890 spectrum, compared to their total absence in the IRD 818 spectrum. Note that when multiple sideband frequencies appear around a bearing defect frequency, they can indicate further bearing deterioration and/or amplitude modulation of BPF1 as the defect passes in and out of the load zone. In the case of the 12-bit spectrum on Figure 34, note the presence not only of a single pair of sidebands, but 2 fully developed sideband families, with a third family on the way. Although this 12-bit spectrum does not indicate imminent failure, particularly due to the absence of any other bearing frequencies or bearing frequency harmonics, the multiple families of sidebands that were totally missed by the 8-bit instrument suggests that this machine deserves close examination.

A similar test was performed on another machine at another time comparing an 8-bit and a 14-bit instrument. In this case, Figure 35 shows the spectrum from an 8-bit collector (SKF/Palomar 6100A), while Figure 36 shows the plot produced by a 14-bit collector (SKF/Palomar 6101A),

8 BIT VERSUS 12 BIT INSTRUMENTS

DECIBEL CONVERSION TABLE

$$dB = 20 \log \frac{V_i}{V_o}$$

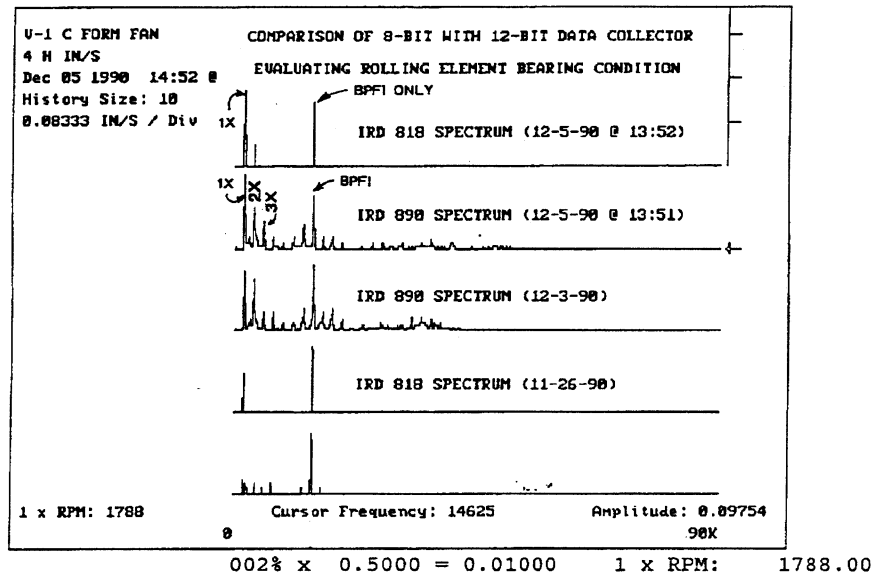
$$\frac{V_i}{V_o} = \text{ANTILOG} \left(\frac{dB}{20} \right)$$

dB	Ratio	Dynamic Range			
		A/D # BITS used	# of bits Value	Largest Number	dB Range
100	100,000:1	1	1	2	6
90	31,623:1	2	2	4	12
80	10,000:1	3	4	8	18
70	3,162:1	4	8	16	24
60	1,000:1	5	16	32	30
50	316:1	6	32	64	36
48	251:1	7	64	128	42
46	200:1	8	128	256	48
40	100:1	9	256	512	54
34	50:1	10	512	1024	60
26	20:1	11	1024	2048	66
20	10:1	12	2048	4096	72
14	5:1	13	4096	8192	78
12	4:1	14	8192	16,384	84
10	3.16:1	15	16,384	32,768	90
8	2.51:1	16	32,768	65,536	96
6	2:1				
5	1.78:1				
4	1.58:1				
3	1.41:1				
2	1.26:1				
1	1.12:1				
0	1:1				

$\frac{4096}{256} = 16 \text{ (24 dB)}$
12 BIT A/D INSTRUMENTS ARE 16 TIMES AS SENSITIVE AS ARE 8 BIT INSTRUMENTS!

1 Part in 100 = 40dB
 1 Part in 200 = 46dB (8 BIT)
 1 Part in 1000 = 60dB
 1 Part in 2000 = 66dB
 1 Part in 4000 = 72dB (12 BIT)

FIGURE 33
CONVERSION CHART FOR NUMBER OF BITS, dB, AND LINEAR
RATIO TO SHOW THE DIFFERENCE BETWEEN 8-BIT AND
12-BIT INSTRUMENTS IN DYNAMIC RANGE



Machine	Pos	Dir	Units	Date	
V-1 C FORM FAN	4	H	IN/S	Dec 05 1990	
Frequency	1 x RPM	Amplitude	Frequency	1 x RPM	Amplitude
1800	1.01	0.14249	40050	22.40	0.01001
2700	1.51	0.02002			
3600	2.01	0.07752			
5400	3.02	0.04996			
7200	4.03	0.02248			
9000	5.03	0.01001			
11025	6.17	0.02002			
12825	7.17	0.04750			
14625	8.18	0.09754			
16425	9.19	0.02248			
18225	10.19	0.02248			
20025	11.20	0.01247			
25650	14.35	0.01001			
27450	15.35	0.01001			
30825	17.24	0.01247			
34425	19.25	0.01247			
36225	20.26	0.01501			

BPFI - 2X

BPFI - 1X

BPFI

BPFI + 1X

BPFI + 2X

BPFI + 3X

DETECTED
BY
12-BIT
IRD 890

T - Threshold adjust Q - Quit

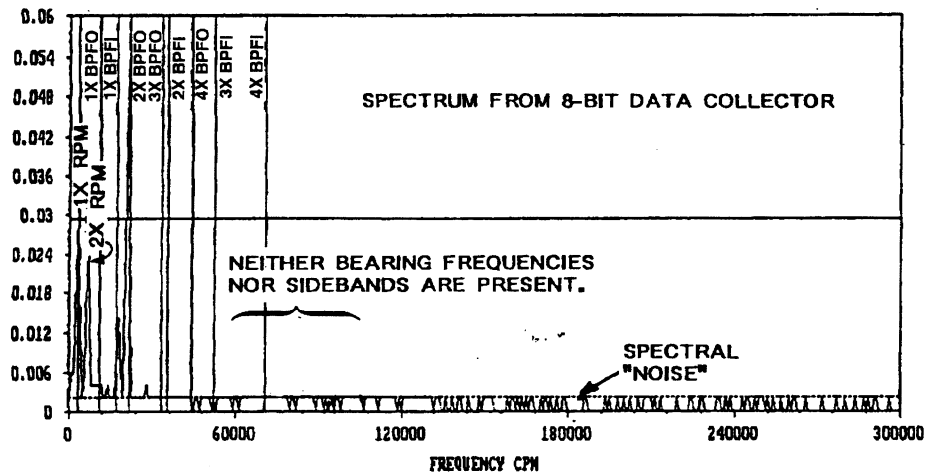
FIGURE 34
COMPARISON OF AN 8-BIT WITH A 12-BIT DATA COLLECTOR TO
EVALUATE ROLLING ELEMENT BEARING CONDITION

SINGLE SPECTRUM PLOT

SET: M-806 TYPE: FFT
 POINT ID: M-806 MT-OBVV DESC:
 WINDOW: HANNING LINES: 400 AVER: 3 FREQ: 0 - 300000 CPM
 DETECT: PEAK RPM: 3575 THRESHOLD: 0.0020 UNITS: IPS

FREQ: 3750.00 AMP: 0.0292 ORDER: 1.049 DEG: ---

SPECTRUM FROM SKF/PALOMAR 6100A (8-BIT A/D)



IDENTIFICATION OF SPECTRAL PEAKS ABOVE THRESHOLD						
NO.	AMP.	FREQ.	ORDER	NO.	AMP.	FREQ.
1.	0.0293	3750.0	1.049			
2.	0.0234	7500.0	2.098			
3.	0.0039	12000.0	3.357			
4.	0.0039	14250.0	3.986			
5.	0.0137	18000.0	5.035			
6.	0.0527	21000.0	5.874			
7.	0.0039	27750.0	7.762			

SPECTRAL ENERGY SUMMARY			
OVERALL	SYNC	SUBSYNC	NONSYNC
0.0733	0.06888	0.005751	0.0244

IDENTIFICATION OF FREQUENCY DATA			
ID	LABEL	AMPL	FREQ
CUSTOM NTN 6314Z BPFO	1*BPFO	0.0039	11250.00
CUSTOM NTN 6314Z BPFI	1*BPFI	0.0078	17250.00
CUSTOM NTN 6314Z BSF	1*BSF	0.0234	7500.00
CUSTOM NTN 6314Z FTF	1*FTF	0.0059	1500.00
CUSTOM NTN 6314Z BPFO	2*BPFO	0.0332	21750.00
CUSTOM NTN 6314Z BPFI	2*BPFI	0.0020	35250.00
CUSTOM NTN 6314Z BSF	2*BSF	0.0020	15000.00
CUSTOM NTN 6314Z FTF	2*FTF	0.0215	3000.00
CUSTOM NTN 6314Z BPFO	3*BPFO	0.0020	33000.00
CUSTOM NTN 6314Z BPFI	3*BPFI	0.0020	52500.00
CUSTOM NTN 6314Z BSF	3*BSF	0.0332	21750.00

CUSTOM NTN 6314Z FTF	3*FTF	0.0117	4500.00
CUSTOM NTN 6314Z BPFO	4*BPFO	0.0020	44250.00
CUSTOM NTN 6314Z BPFI	4*BPFI	0.0020	70500.00
CUSTOM NTN 6314Z BSF	4*BSF	0.0020	29250.00
CUSTOM NTN 6314Z FTF	4*FTF	0.0020	5250.00

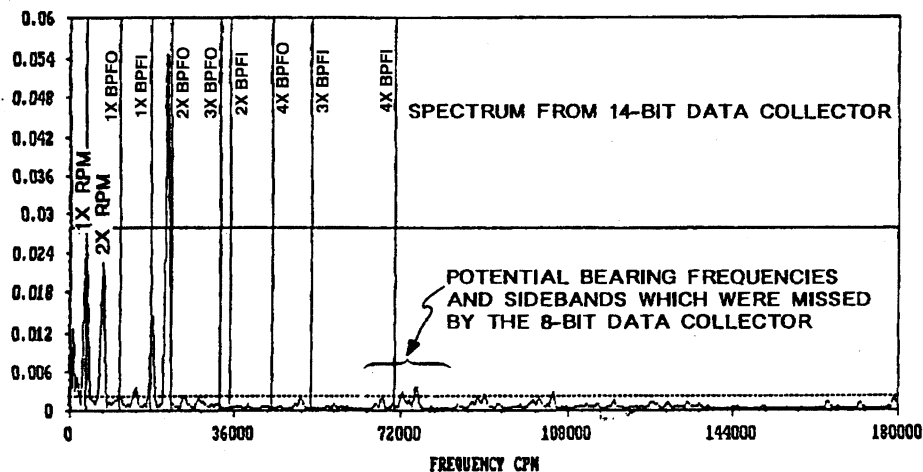
FIGURE 35
 SPECTRUM FROM AN SKF/PALOMAR 6100A (8-BIT A/D)

SINGLE SPECTRUM PLOT

SET: M-806 TYPE: FFT
 POINT ID: M-806 MT-08VV DESC:
 WINDOW: HANNING LINES: 400 AVER: 3 FREQ: 0 - 180000 CPM
 DETECT: PEAK RPM: 3575 THRESHOLD: 0.0020 UNITS: IPS

FREQ: 3600.00 AMP: 0.0277 ORDER: 1.007 DEG: ---

SPECTRUM FROM SKF/PALOMAR 6101A (14-BIT A/D)



IDENTIFICATION OF SPECTRAL PEAKS ABOVE THRESHOLD

NO.	AMP.	FREQ.	ORDER	NO.	AMP.	FREQ.	ORDER
1.	0.0122	900.0	0.252	9.	0.0021	25200.0	7.049
2.	0.0048	1800.0	0.503	10.	0.0020	50400.0	14.098
3.	0.0278	3600.0	1.007	11.	0.0027	72450.0	20.266
4.	0.0221	7200.0	2.014	12.	0.0037	75150.0	21.021
5.	0.0021	11250.0	3.147	13.	0.0020	88650.0	24.797
6.	0.0033	14400.0	4.028	14.	0.0022	90000.0	25.175
7.	0.0139	18000.0	5.035	15.	0.0026	104850.0	29.329
8.	0.0541	21150.0	5.916				

SPECTRAL ENERGY SUMMARY			
OVERALL	SYNC	SUBSYNC	NONSYNC
0.0687	0.06293	0.0111	0.02523

IDENTIFICATION OF FREQUENCY DATA

ID	LABEL	AMPL	FREQ
CUSTOM NTN 6314Z BPFO	1*BPFO	0.0019	10800.00
CUSTOM NTN 6314Z BPFI	1*BPFI	0.0105	17550.00
CUSTOM NTN 6314Z BSF	1*BSF	0.0221	7200.00
CUSTOM NTN 6314Z FTF	1*FTF	0.0029	1350.00
CUSTOM NTN 6314Z BPFO	2*BPFO	0.0014	22050.00
CUSTOM NTN 6314Z BPFI	2*BPFI	0.0002	35100.00
CUSTOM NTN 6314Z BSF	2*BSF	0.0014	14850.00
CUSTOM NTN 6314Z FTF	2*FTF	0.0009	2700.00
CUSTOM NTN 6314Z BPFO	3*BPFO	0.0004	32850.00
CUSTOM NTN 6314Z BPFI	3*BPFI	0.0002	52650.00

CUSTOM NTN 6314Z BSF	3*BSF	0.0014	22050.00
CUSTOM NTN 6314Z FTF	3*FTF	0.0121	4050.00
CUSTOM NTN 6314Z BPFO	4*BPFO	0.0001	44100.00
CUSTOM NTN 6314Z BPFI	4*BPFI	0.0004	70650.00
CUSTOM NTN 6314Z BSF	4*BSF	0.0011	29250.00
CUSTOM NTN 6314Z FTF	4*FTF	0.0011	5400.00

FIGURE 36
 SPECTRUM FROM AN SKF/PALOMAR 6101A (14-BIT A/D)

again connecting the identical accelerometer to each analyzer. Note from Figure 36 that 14-bit A/D instruments have a theoretical dynamic range of 84 dB which would make them 64 times the amplitude sensitivity of 8-bit models (SKF/Palomar specifications show an 80 dB dynamic range which would make the 6101A approximately 40X as sensitive as its 6100A counterpart). Looking at Figure 35 note particularly the bearing frequency harmonic and accompanying sidebands at about 72,000 CPM in Figure 36 (captured by the 14-bit collector) which again are totally absent from the 8-bit spectrum of Figure 35. In addition, note the elevated "noise floor" of the 8-bit spectrum which appeared at about .0025 in/sec when this spectrum was set to the same full scale level of .06 in/sec as was the 14-bit spectrum. Importantly, it has been personally witnessed that bearings have had considerable damage even though they had defect frequency amplitudes even lower than this .0025 in/sec when they had a number of bearing frequency harmonics accompanied by multiple sideband families present in their spectra.

Therefore, it appears that 8-bit data collectors having a dynamic range on the order of 46 to 48 dB are inadequate for dependably detecting potentially serious bearing, gear, electrical and hydraulic flow problems in today's industrial machinery and machine tools due to their amplitude insensitivity/dynamic range problems, particularly where high amplitudes at 1X or 2X RPM coexist with low level (but important) frequencies and sidebands. An analyst should instead employ FFT instruments with a minimum of 12 A/D bits and a 70 to 72 dB dynamic range.

3.8 WHAT IS OVERALL VIBRATION?

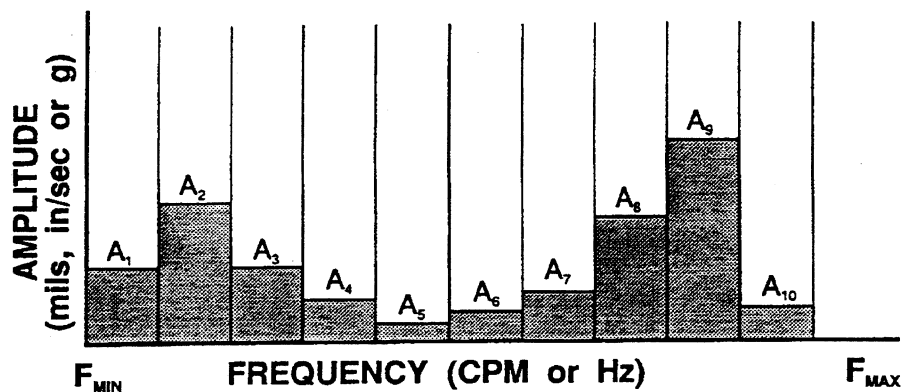
Overall vibration is different from vibration at specific frequencies as discussed thus far in that it is a total summation of all the vibration, with no regard to any particular frequency. There are two types of overall vibration levels that an analyst can decide on using: DIGITAL Overall Level and ANALOG Overall Level. The Analog Overall Level is limited by the vibration instrument and transducer performance characteristics used, whereas the Digital Overall Level is limited primarily by the frequency range (0 - F_{MAX}) specified by the analyst. The advantage of specifying an Analog Overall option over a Digital Overall will be discussed in this section. To begin with, the Digital Overall Level will be considered first.

3.81 DIGITAL (OR SPECTRAL) OVERALL LEVEL

The vibration seen in a spectrum is the total sum of all the vibration measured by the transducer within the frequency span chosen. The transducer may even sense vibration from an adjacent machine and include it within the vibration spectrum. Whatever is "seen" by the transducer, regardless of its source, becomes part of the spectrum and the overall vibration.

On the latest equipment, this spectral overall is calculated automatically either within the analyzer and/or back within the host software of the computer. However, not every analyst is equipped with the latest equipment. Older instruments and software often do not have this capability. Nevertheless, there are two techniques that an analyst can use to determine the spectral or "Digital Overall". One technique employs an exact equation and the other technique uses an approximation formula with a correction factor. Figure 37 shows the exact equation that can be used for calculating the digital overall vibration level of a spectrum. This is known as the "RSS" level, meaning "Root Sum Square" level as defined in the equation on Figure 37.

To calculate the exact Digital Overall Vibration Level of a spectrum, take the amplitude of each frequency bin (A_i) and square it (A_i^2), add all the squared amplitudes together; take the square root of that sum; and divide this sum by the noise factor for the FFT Window chosen (the Hanning Window is used most for PMP route measurements and has a 1.5 noise factor). Of course, this is a very lengthy process for a 400-line spectrum (and even more so for a 3200-line FFT). However, this merely illustrates what the latest equipment will do automatically; it is rarely necessary to do it manually today.



$$OA_{RSS} = \frac{\sqrt{\sum_{i=1}^n A_i^2}}{\sqrt{N_{BF}}} = \frac{\sqrt{A_1^2 + A_2^2 + A_3^2 + \dots + A_n^2}}{\sqrt{1.5}}$$

OA_{RSS} = Digital Overall Level of Vibration Spectrum.

RSS = Root Sum Square Level which is the "Summation of Energy" in the spectrum from F_{MIN} to F_{MAX} .

n = Number of FFT Lines of Resolution.

A_i = Amplitude of Each of the FFT Lines.

N_{BF} = Noise Bandwidth for Window Chosen 1.5 for Hanning Window.

Which means $OA_{RSS} = .8165 \sqrt{\sum_{i=1}^n A_i^2}$

**FIGURE 37
EXACT EQUATION FOR CALCULATING DIGITAL OVERALL LEVEL
OF A SPECTRUM**

Figure 38 shows an approximate formula for calculating the overall vibration spectrum and a sample spectrum showing how this approximation is applied. Here, the amplitudes from 5 separate frequency peaks were combined to compute the approximate overall level of the spectrum. Figure 38 also illustrates a problem with relying on an increase in overall vibration levels to indicate a worsening condition; in this case, a deteriorating bearing. The baseline spectrum showed that a bearing frequency had an amplitude of .030 in/sec. However, four weeks later, the spectrum taken showed that the bearing frequency peak had increased to .120 in/sec. The peak was now 4X larger. Notice the calculated overall vibration levels that were determined from the approximate equation. At baseline, the overall vibration level was .435 in/sec. Four weeks later, the overall vibration was calculated to be .450 in/sec. If one would have only relied on overall vibration levels, this slight increase in value would be no cause for alarm. However, there is cause for alarm. A spectral peak increased 4 times in value in just 4 weeks! The point is, whether the overall vibration level is calculated or measured is not significant; it will not adequately indicate a change in spectral information. A spectrum will show minute, as well as large changes in the spectral peaks and/or the amplitudes.

Figure 39 is a sample of an estimated overall vibration calculation using an actual spectrum. Note that the estimated overall value that was calculated is .161 in/sec. As a comparison, the actual overall, as measured by the data collector, was .182 in/sec. The actual overall is always higher than the estimated method since it includes all the individual amplitudes within each bin, not only the detectable peaks over the user-defined threshold. (An example of a user-defined threshold can be seen in Figure 16). By multiplying the estimated value by a 1.1 correction factor, derived experimentally, the estimated vibration will be fairly close to the actual (i.e., $.161 \times 1.1 = .177$ in/sec; very close to the actual .182 in/sec overall vibration). Fortunately, the data collectors and computer software perform all these calculations and one only needs to do this calculation under unusual circumstances.

3.82 ANALOG OVERALL LEVEL

One of the problems of looking at only the Digital (or Spectral) Overall is that any significant vibration occurring outside the frequency range ($0 - F_{MAX}$) specified by the analyst will not be "seen". For example, if a high level of .60 in/sec was occurring out at a frequency of 100,000 CPM and an F_{MAX} of only 60,000 CPM was chosen, this high-amplitude 100,000 CPM frequency would not be included within the Digital Overall. In this case, the Digital Overall might only be calculated as .20 in/sec. This would be far below the Digital Overall value that could be obtained if a frequency range out to 120,000 CPM had been chosen.

To overcome this problem, some PMP systems determine the overall by looking directly at the time waveform over a very wide fixed frequency range. Then, the overall measured by the analyzer will be totally independent of any frequency-span spectrum chosen by the user. This is what is known as the Analog Overall. For example, one system uses a time waveform having a fixed frequency span from 300 to 3,900,000 CPM (5 to 65,000 Hz). In the above example (with the .60 in/sec at 100,000 CPM), the Analog Overall would be dramatically higher than the Spectrum/Digital Overall where an F_{MAX} of only 60,000 CPM was chosen. For this reason, if the PMP system allows the option of finding either the Spectrum (or Digital) Overall or Analog Overall, choose the Analog Overall option.

By comparing the differences in value between the Analog Overall and the Digital Overall, the analyst will be able to determine if there is more spectral content that is beyond the F_{MAX} than was specified. If there is, he can rectify the situation by specifying a larger F_{MAX} (perhaps to the maximum) in order to capture that data. Regardless of which Overall Level measuring system is used, a spectrum is a better indicator of the actual change in machine condition. The Analog Overall Level merely lets the analyst know if his F_{MAX} is set high enough to capture all the spectral content.

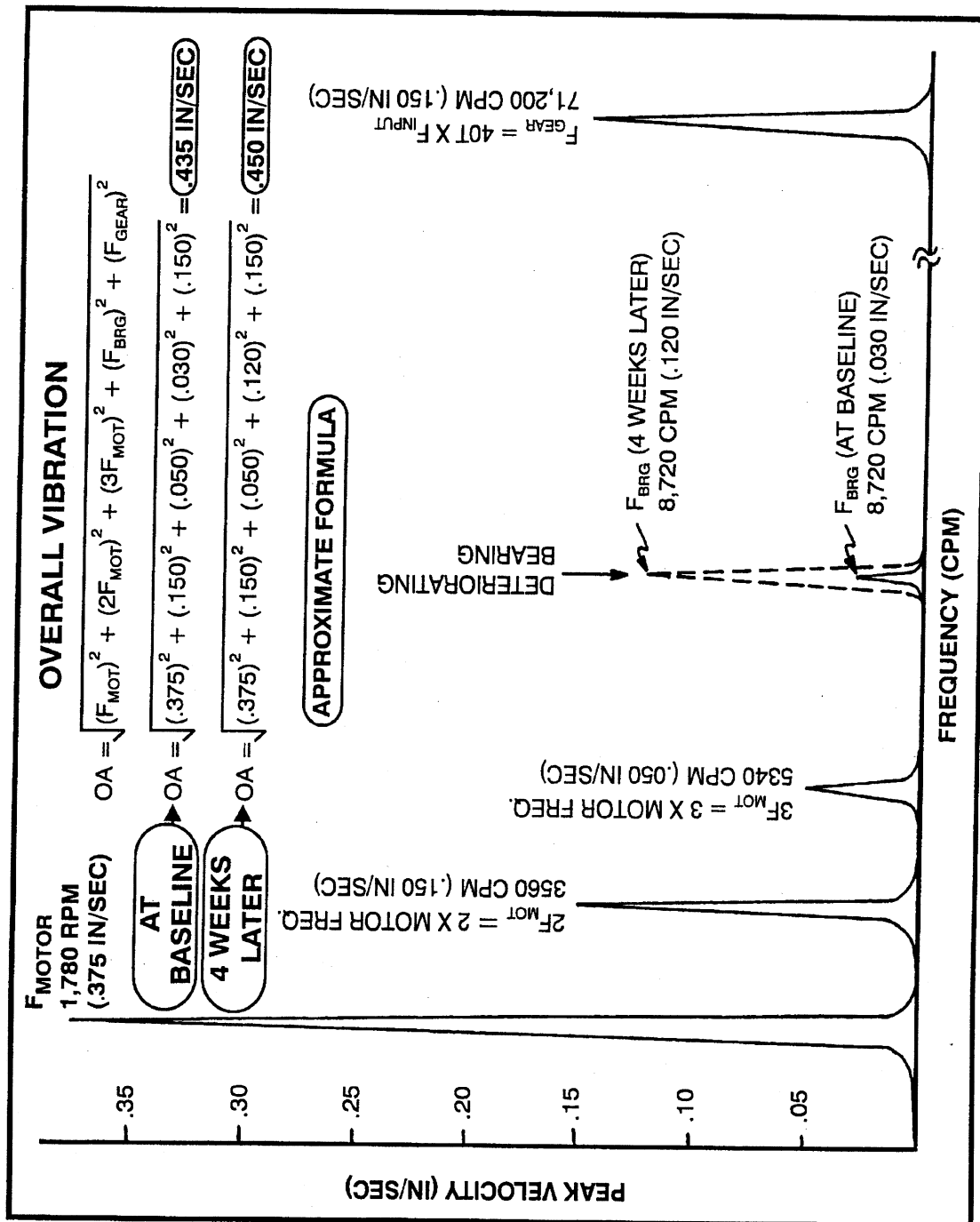
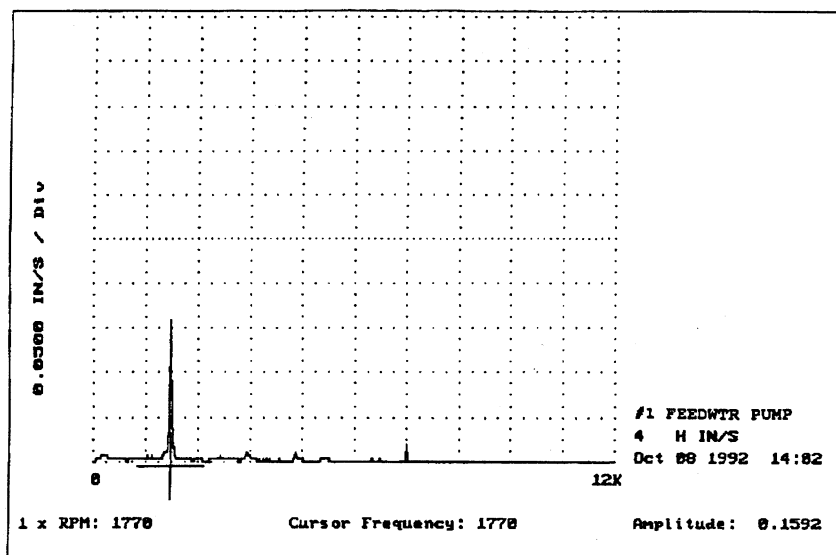


FIGURE 38
APPROXIMATE FORMULA FOR CALCULATING THE OVERALL LEVEL
OF A SPECTRUM



1 1/2 x 0.500 = 0.0050 1 x RPM: 1770.00

Machine	Pos	Dir	Units	Date
#1 FEEDWTR PUMP	4	H	IN/S	Oct 08 1992 14:02
Frequency	1 x RPM	Amplitude		
240	0.136	0.0057	where OA = Overall Level of Vibration n = Number of FFT Lines of Resolution Ai = Amplitude of Each of the FFT Lines	
300	0.169	0.0071		
1230	0.695	0.0051		
1770	1.000	0.1592		
2910	1.644	0.0050		
3510	1.983	0.0106		
4650	2.627	0.0079		
7200	4.068	0.0201		

$$OA = \sqrt{\sum_{i=1}^n A_i^2}$$

$$OA = \sqrt{(A_1)^2 + (A_2)^2 + (A_3)^2 + (A_4)^2 + (A_5)^2 + (A_6)^2 + (A_7)^2 + (A_8)^2}$$

$$OA = \sqrt{(.0057)^2 + (.0071)^2 + (.0051)^2 + (.1592)^2 + (.0050)^2 + (.0106)^2 + (.0079)^2 + (.0201)^2}$$

$$= \sqrt{.026}$$

$$OA = .161 \text{ inch/sec}$$

$$OA = .161 \times 1.1 \text{ (CORRECTION FACTOR)}$$

$$OA = .177 \text{ inch/sec}$$

T - Threshold adjust Q - Quit

FIGURE 39
SAMPLE CALCULATION OF APPROXIMATE OVERALL (OA) VIBRATION

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